CC BY

IT05: Robotics and Automatic Control #1
Feedback and PID control
フィードバックとPID制御

Prof. Keitaro Naruse

naruse@u-aizu.ac.jp

Office: 322B

CC BY

What You Will Learn in This Course
この科目で学ぶこと

Automatic control /自動制御
- Classical control theory (PID control)
  古典制御理論（PID制御）
- DC motor speed control (Vehicle)
  DCモータの速度制御（車両）
- DC motor position control（Arm）
  DCモータの位置制御（アーム）

Lecture: theory
講義: 理論

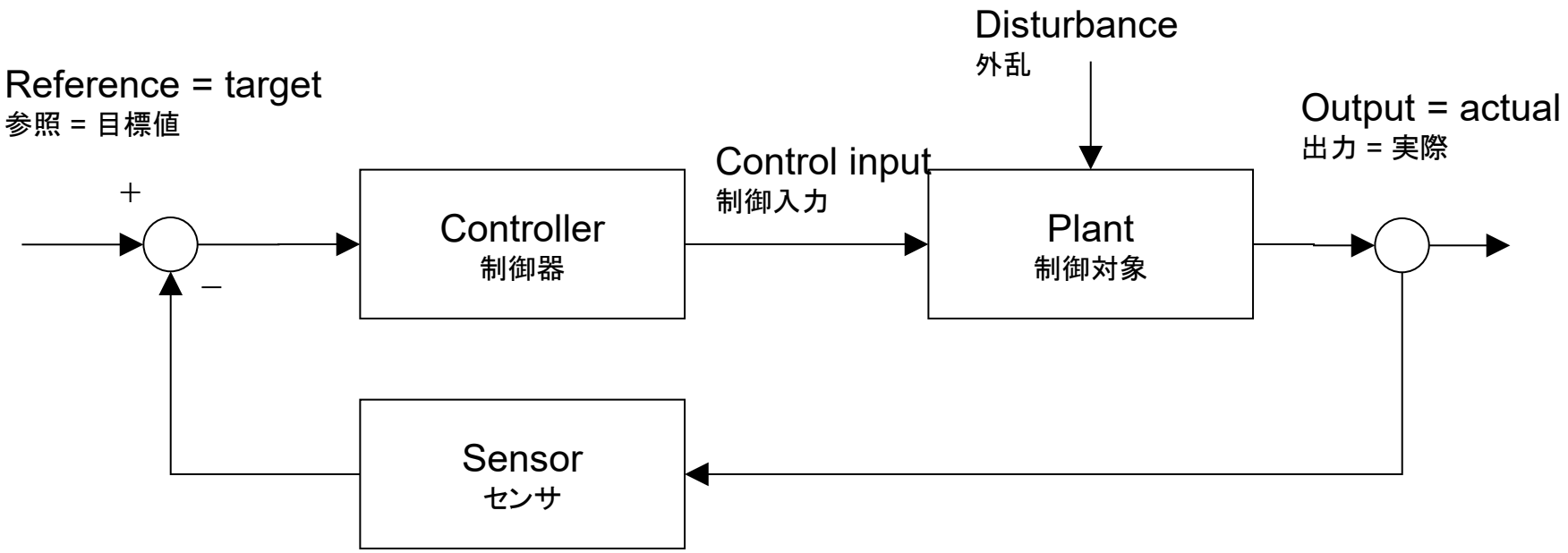
Robotics / ロボット工学
- Vehicle kinematics (wheel velocity to vehicle one)
  車両の運動学（車輪速度から車体速度）
- Arm kinematics (joint angular velocity to hand velocity)
  アームの運動学（関節速度から手先速度）
- World modeling methods
  世界のモデル化手法
- Path planning method
  経路計画手法

Exercise with matlab
演習: matlab

Laplace Transform, Transfer Function, and Block Diagrams

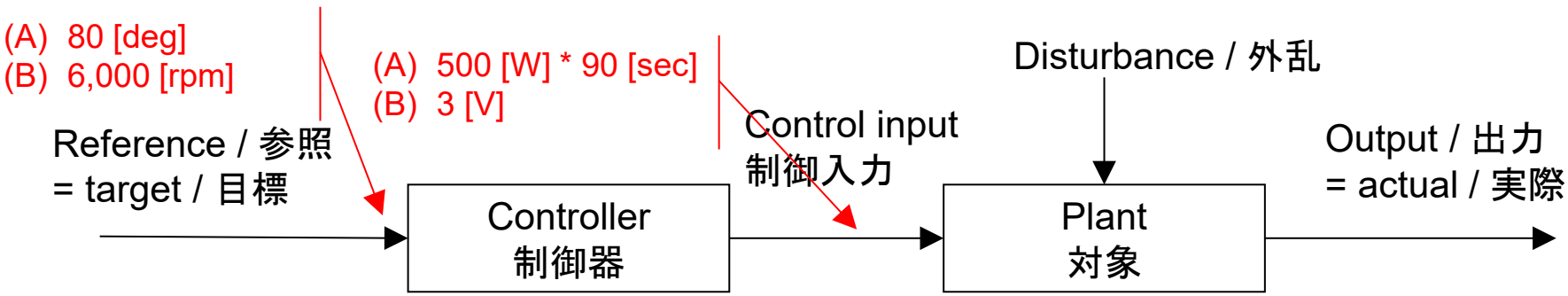
ラプラス変換，伝達関数，ブロック線図

Feedback Control
フィードバック制御



CC BY

Concept of Feedforward Control
フィードフォワード制御の概念



Example of Feedforward control / フィードフォワード制御の例

(A) Suppose heating water of 20 deg. to 80 deg. by microwave.
If we know an amount of water, we can find hearing time.
電子レンジで20度の水を加熱して80度のお湯にすることを考える
水の分量が分かっていたら，加熱時間を事前に求めることができる
180 [ml]: 500[W] * 90 [sec]

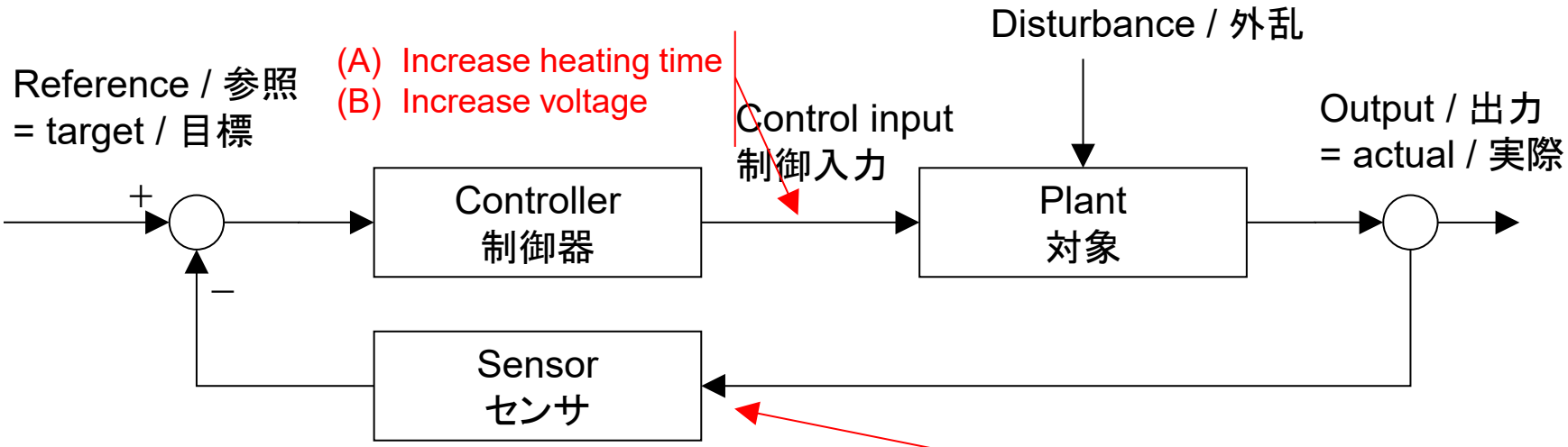
How long should we heat it if we do not know water volume?
水の量が分からないときは何秒加熱する？

(B) Suppose rotating a DC motor at 6,000 rpm.
If we know motor specs and load, we can find an applied voltage
DCモータを毎分6,000回転で回すことを考える.
モータの性能と負荷が分かっていらた，印加電圧を求めることができる

What voltage should we apply f we do not know load?
負荷が分からないときは何ボルト印加する？

CC BY

Concept of Feedback Control
フィードバック制御の概念



Idea of Feedback control / フィードバック制御のアイディア

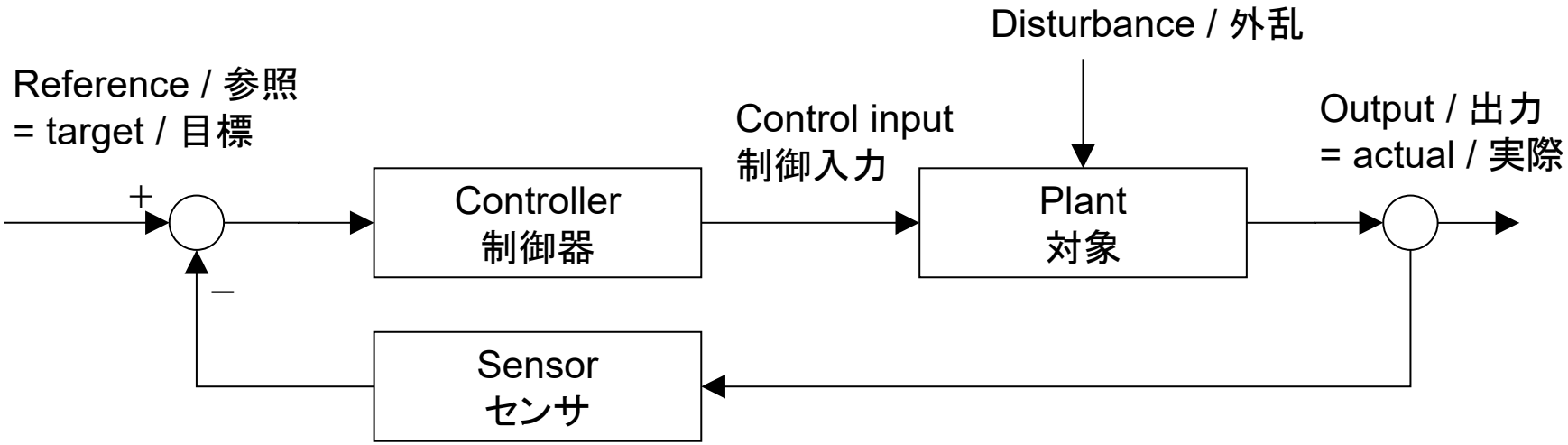
(A) 80 [deg]
(B) 6,000 [rpm]

(A) Suppose heating water to 80 deg. by microwave. If we can measure a temp., we can heat it until it comes to target temp.
温度計測が可能なら，目標温度になるまで加熱すればよい

(B) Suppose rotating a DC motor at 6,000 rpm. If we can measure a motor speed, we increase an applied voltage until it comes to a target speed.
DCモータを毎分6,000回転で回すことを考える．モータの回転速度を計測できるのなら，目標速度になるまで印加電圧を増加させる

CC BY

Theory of Feedback Control
フィードバック制御理論

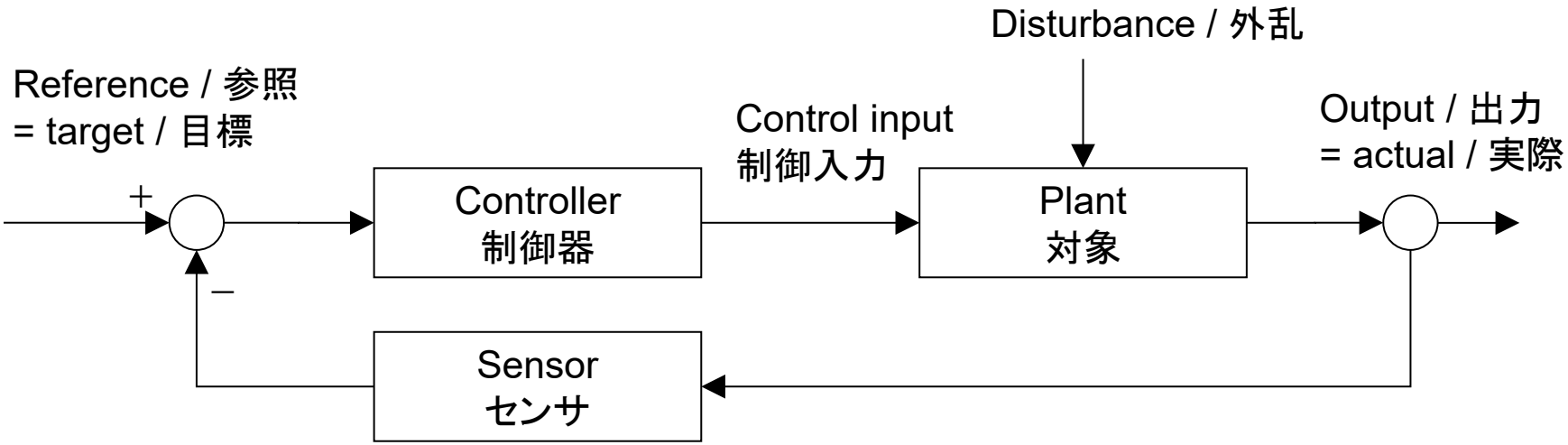


The objective of this part: この部分の目的

- Represent a feedback control system mathematically (transfer functions and block diagrams) フィードバック制御システムを数学的に表現する（伝達関数とブロック線図）
- Discuss stability and error of a system システムの安定性と誤差を議論する
- Design a controller by control specifications 制御目標に応じで制御器を設計する
- Evaluate a developed system (Simulation by matlab) 開発されたシステムを評価する

CC BY

Transfer Function and Block Diagram
伝達関数とブロック線図



- Each of the blocks can be considered as a signal transducer between input and output.
  上の各ブロックは入力と出力の間の信号変換器とみなすことができる
- In classical control theory, it is represented by a transfer function, and a whole system is done by a block diagram.
  古典制御理論では，これらを伝達関数で表現し，システム全体はブロック線図と呼ばれる

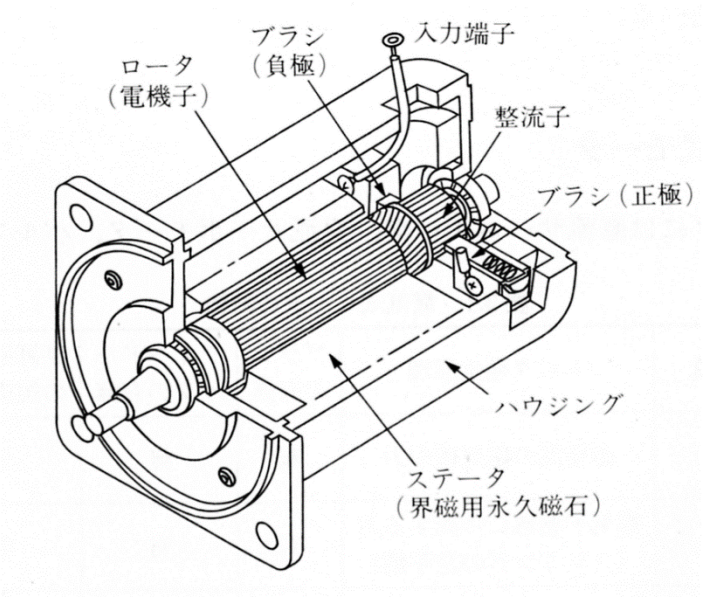
CC BY

Transfer Function and Laplace Transform
伝達関数とラプラス変換

Applied voltage [V]
電圧

→ DC motor →

Angular velocity [rad/s] [rpm]
回転速度



https://en.wikipedia.org/wiki/DC_motor

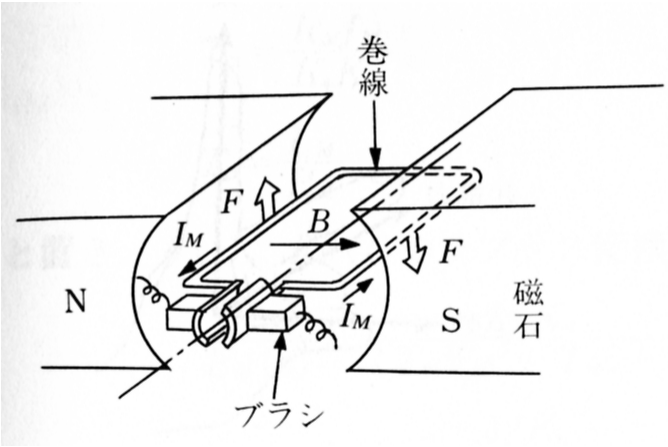
DC motor is a signal transducer represented by a set of equations
複数の方程式で表現できる信号変換器

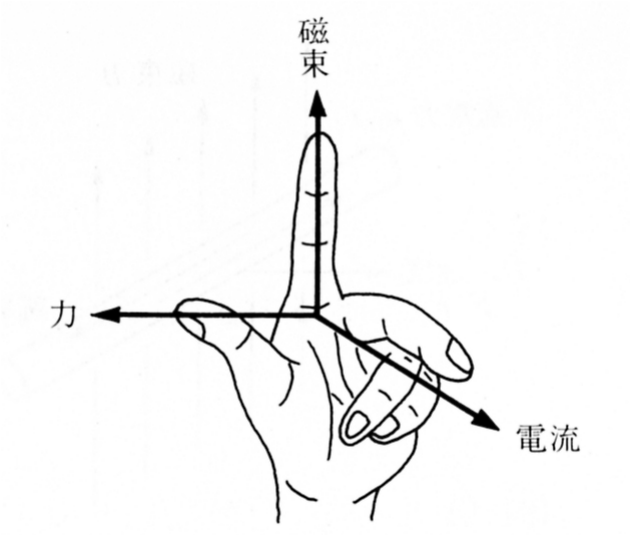
- As actuator アクチュエータ
  Transfer from applied voltage > current > Lorenz force > torque
  電圧 > 電流 > ローレンツ力 > トルク
- As generator 発電機
  Transfer from torque > Lorenz force > current and voltage
  トルク > ローレンツ力 > 電流と電圧

CC BY

DC Motor Equation (1/4)

DCモータの方程式（1/4）





Current -> Torque

$$F(t) = BLI_M(t)$$

F: Force [N]
B: Magnetic Flux Density [T]
L: Coil length [m]
I_M: Current [A]

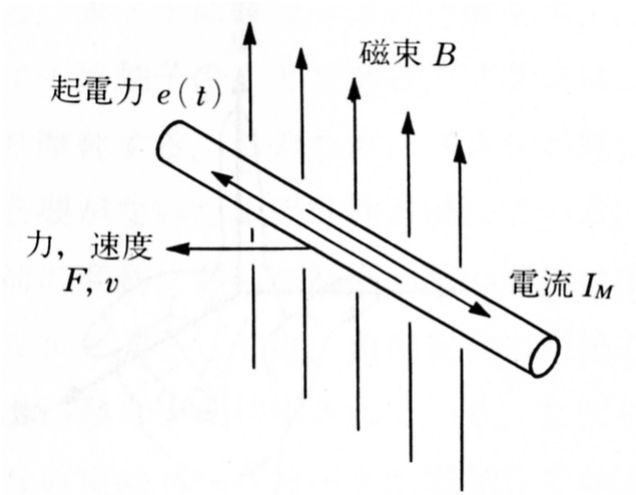
$$\tau(t) = rF(t)$$
$$= K_T I_M(t) \quad K_T = rBL$$

τ: Torque [Nm]
r: Radius [m]
K_T: Torque constant [NmA^{-1}]

CC BY

DC Motor Equation (2/4)

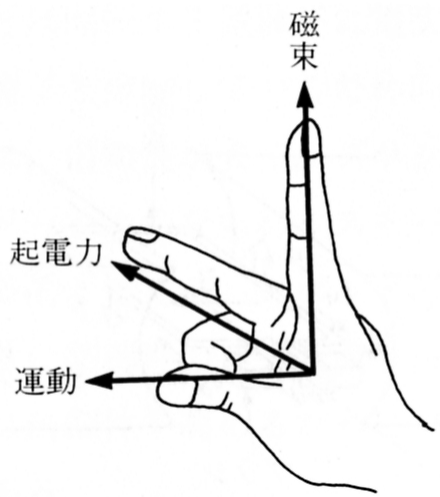
DCモータの方程式（2/4）



Revolution -> Voltage
Back EMF (ElectroMotive Force)

$$e(t) = BLv(t)$$

e: Voltage [V]
B: Magnetic Flux Density [T]
L: Coil length [m]
v: velocity [m/s]



$$e(t) = K_e \dot{\theta}(t) \qquad K_E = rBL \qquad \dot{\theta}(t) = \frac{v(t)}{r}$$

K_e: BackEMG const. [Vs/rad]
θ: rotar angle [rad]
Note: $K_T = K_E$

CC BY

DC Motor Equation (3/4)

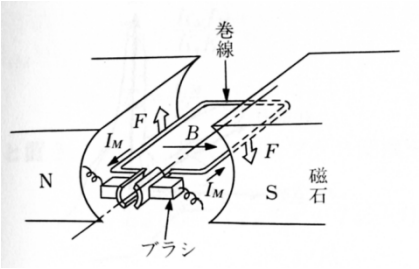
DCモータの方程式（3/4）

Make an equation on voltage:

Motor: Resist EMF + BackEMF + Induced EMF

E_M

$$E_M = R_M I_M + K_E \dot{\theta}(t) + L_M \frac{dI_M(t)}{dt}$$



E_M: Applied Voltage [V]
R_M: Resistance [Ω]
I_M: Current [A]

K_E: Back EMF const. [Vs/rad.]
L_M: Coil Inductance [H]
θ: Rotar angle [rad.]

CC BY

DC Motor Equation (4/4)

DCモータの方程式（4/4）

Generated torque accelerates a rotar

$$\tau(t) = J_M \ddot{\theta}(t) + \tau_L$$

High order differential equation, and difficult to solve
高次の微分方程式で解くのが困難

J_M: rotar intertia moment [Nm/rad.^2]
τ_L: load torque [Nm]

If τ_L=0, from the above equations, eliminating τ and I_M,

$$E_M(t) = \left(\frac{L_M J_M}{K_E K_T} \dddot{\theta}(t) + \frac{R_M J_M}{K_E K_T} \ddot{\theta}(t) + \dot{\theta}(t) \right) K_e$$

$$= \left(T_m T_e \dddot{\theta}(t) + T_m \ddot{\theta}(t) + \dot{\theta}(t) \right) K_e$$

$$T_m = \frac{R_M J_M}{K_e K_T} \quad T_e = \frac{L_M}{R_M}$$

T_m: mechanical time const.
T_e: electrical time const.

CC BY

Laplace transform
ラプラス変換

- System model is represented by differential equations because states change by time, but it is difficult to analyze
  システムの状態は時間とともに変化するので，モデルは微分方程式で記述されるが，解析が困難
- Empirically, system outputs follow decaying oscillation such as $e^{-\tau t}\cos(\omega t+\phi)$
  経験的にシステム出力は一般に時間とともに減衰する振動が多い，たとえば $e^{-\tau t}\cos(\omega t+\phi)$
- Introducing Laplace transform, we can represent differential and integral operations by simple ones(addition and multiplication)
  ラプラス変換を導入することにより，微分・積分の操作を簡単な代数的操作（加減乗除）に変換できる

CC BY

Example of Laplace transform
ラプラス変換の例

Motion equation of mass-spring / バネ振子の運動方程式

$$m\ddot{x}(t) = -kx(t) + f(t)$$

f(t): force　　F(s): Laplace transformation of f(t)
x(t): position　　X(s):Laplace transformation of x(t)

$$ms^2 X(s) = -kX(s) + F(s)$$

$$F(s) = \left(ms^2 + k\right)X(s),\ X(s) = \frac{1}{\sqrt{mk}}\frac{\sqrt{k/m}}{\left(s^2 + k/m\right)}F(s)$$

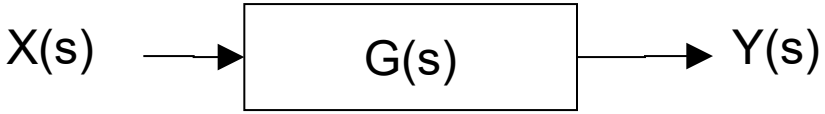
$$x(t) = \frac{1}{\sqrt{mk}}\sin\left(\sqrt{\frac{k}{m}}t\right)f(t)$$

We can solve differential equation easily /
微分方程式を簡単に解くことができる

CC BY

Transfer function
伝達関数

- System representation method
  システムの表現方法
  - Block element: One-input and one-output
    ブロック要素: 1 入力，1 出力
  - Each block is considered as signal transducer
    各要素を信号変換器とみなす
  - Model a block by an output/input ratio of Laplace transformation
    ラプラス変換による出力と入力の比でモデル化する
  - The block element is a transfer function
    そのブロック要素が伝達関数



$$G(s) = \frac{Y(s)}{X(s)}$$

CC BY

Laplace transformation table
ラプラス変換表

$$f(t) \longleftrightarrow F(s)$$

$$\frac{df(t)}{dt} \longleftrightarrow sF(s) - f(0)$$

$$\int_0^t f(u)du \longleftrightarrow \frac{1}{s}F(s)$$

$$\delta(t) \longleftrightarrow 1$$

$$H(t) \longleftrightarrow \frac{1}{s}$$

Step function

$$t^n \longleftrightarrow \frac{n!}{s^{n+1}}$$

$$\sin(\omega t) \longleftrightarrow \frac{\omega}{s^2 + \omega^2}$$

$$\cos(\omega t) \longleftrightarrow \frac{s}{s^2 + \omega^2}$$

$$t^n f(t) \longleftrightarrow (-1)^n \frac{d^n}{ds^n} F(s)$$

$$e^{-at} f(t) \longleftrightarrow F(s+a)$$

$$e^{-at} \longleftrightarrow \frac{1}{s+a}$$

$t \in \Re$ Time domain/ 時間領域 $\longleftrightarrow$ Complex domain / 複素領域 $s = \sigma + j\omega$

CC BY

Definition of Laplace transform
ラプラス変換の定義

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

Convert a time function of f(t) to a complex one F(s)
時間領域の関数 *f*(*t*) を複素領域の関数 F(s) に変換する

After manipulations in a complex domain, transform back to a time domain /
様々な計算の後，時間領域に逆変換する

$$f(t) = \lim_{p \to \infty} \frac{1}{2\pi j} \int_{c-jp}^{c+jp} F(s) e^{st} ds$$

$F(s) = L[f(t)]$ Laplace transform / ラプラス変換

$f(t) = L^{-1}[F(s)]$ Inverse Laplace transform / 逆ラプラス変換

CC BY

Properties of Laplace transform
ラプラス変換の性質

Representation of transient and oscillating phenomena
過渡的かつ振動的な現象を表現

$$e^{-st} = e^{-(\delta + j\omega)t} = e^{-\delta t}\left(\cos(-\omega t) + j\sin(-\omega t)\right)$$

Linearity
線形性

$$L\left[af(t) + bg(t)\right] = aF(s) + bG(s)$$

$$L^{-1}\left[aF(s) + bG(s)\right] = af(t) + bg(t)$$

Convolution
畳みこみ

$$L\left[f(t) * g(t)\right] = F(s)G(s)$$

$$L^{-1}\left[F(s)G(s)\right] = f(t) * g(t)$$

CC BY

Properties of Laplace transform
ラプラス変換の性質

By setting f(0) = 0, differential and integral operations can be represented by multiplication and division
f(0) = 0 となるように設定することに微分と積分が掛け算と割り算で表現可能

$$L\left[\frac{df(t)}{dt}\right] = sF(s) - f(0) \approx sF(s) \qquad L\left[\frac{d^n f(t)}{dt^n}\right] \approx s^n F(s)$$

Differential
微分

$$L\left[\frac{d^2 f(t)}{dt^2}\right] = s^2 F(s) + sf(0) - f'(0) \approx s^2 F(s)$$

Integral
積分

$$L\left[\int_0^t f(u)du\right] = \frac{1}{s}F(s)$$

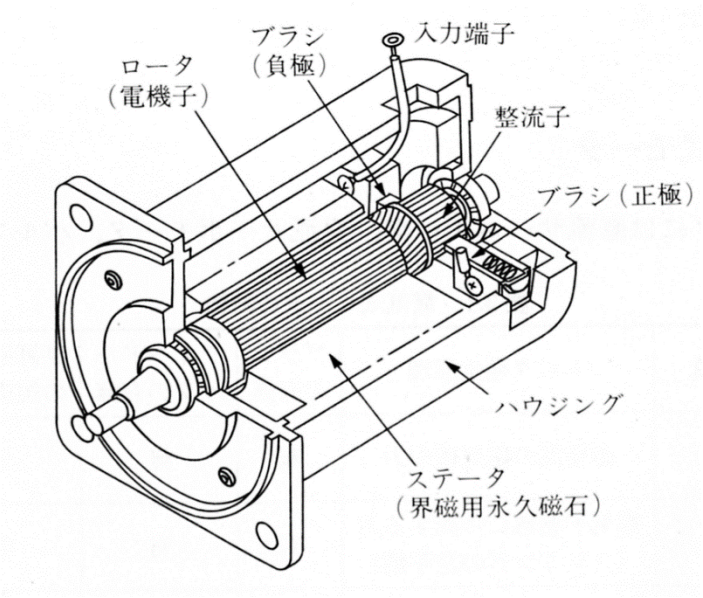
CC BY

Transfer Function and Laplace Transform
伝達関数とラプラス変換

Applied voltage [V]
電圧

→ DC motor →

Angular velocity [rad/s] [rpm]
回転速度



DC motor is a signal transducer represented by a set of equations
複数の方程式で表現できる信号変換器

- As actuator アクチュエータ
  Transfer from applied voltage > current > Lorenz force > torque
  電圧 > 電流 > ローレンツ力 > トルク
- As generator 発電機
  Transfer from torque > Lorenz force > current and voltage
  トルク > ローレンツ力 > 電流と電圧

https://en.wikipedia.org/wiki/DC_motor

CC BY

Transfer Function of DC Motor
DCモータの伝達関数

Differential equation of DC motor / DCモータの微分方程式

$$
\begin{aligned}
e_M(t) &= \left(\frac{L_M J_M}{K_E K_T} \dddot{\theta}(t) + \frac{R_M J_M}{K_E K_T} \ddot{\theta}(t) + \dot{\theta}(t) \right) K_e \\
&= \left(T_m T_e \dddot{\theta}(t) + T_m \ddot{\theta}(t) + \dot{\theta}(t) \right) K_e \\
&= \left(T_m T_e \ddot{\omega}(t) + T_m \dot{\omega}(t) + \omega(t) \right) K_e
\end{aligned}
$$

$e_M(t)$: Applied voltage to DC motor
$\omega(t)$: Angular velocity of DC motor

Laplace transformed differential equation of DC motor
ラプラス変換されたDCモータの微分方程式

$$
\begin{aligned}
E_M(s) &= \left(T_m T_e s^2 \Omega(s) + T_m s \Omega(s) + \Omega(s) \right) K_e \\
&= \left(T_m T_e s^2 + T_m s + 1 \right) K_e \Omega(s)
\end{aligned}
$$

$E_M(s)$: Laplace transformed applied voltage
$\Omega(s)$: Laplace transformed angular velocity

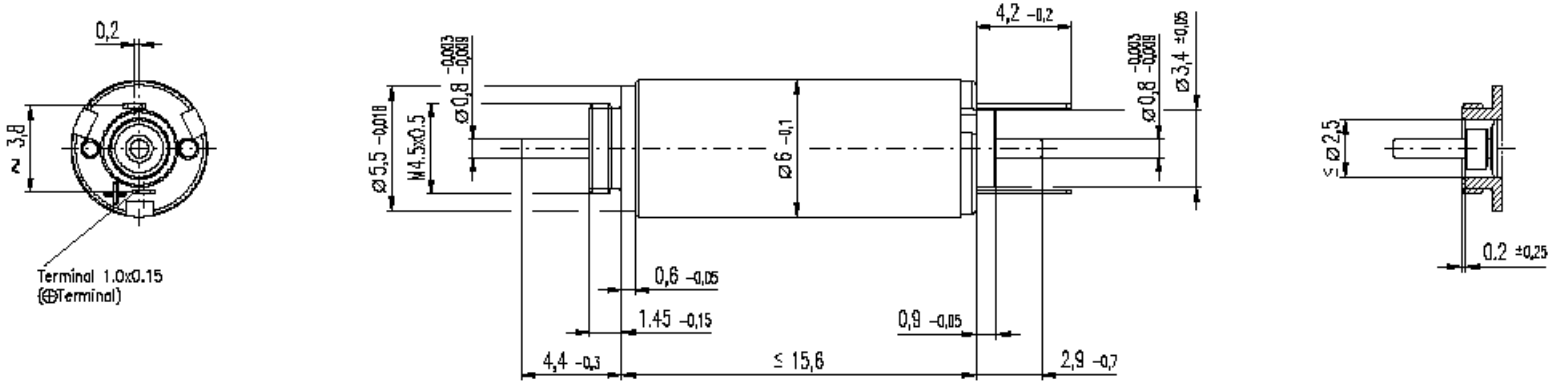
$$
G(s) = \frac{\Omega(s)}{E_M(s)} = \frac{1}{\left(T_m T_e s^2 + T_m s + 1 \right) K_e}
$$

Transfer function of DC motor
from applied voltage to angular velocity
印加電圧から回転速度への伝達関数

Motor Spec. Sheet (1/2)

RE 6 ∅6 mm, 貴金属ブラシ, 0.3 Watt

NEW



Maxon motor

CC BY

Motor Spec. Sheet (2/2)

R_m

K_T

K_E

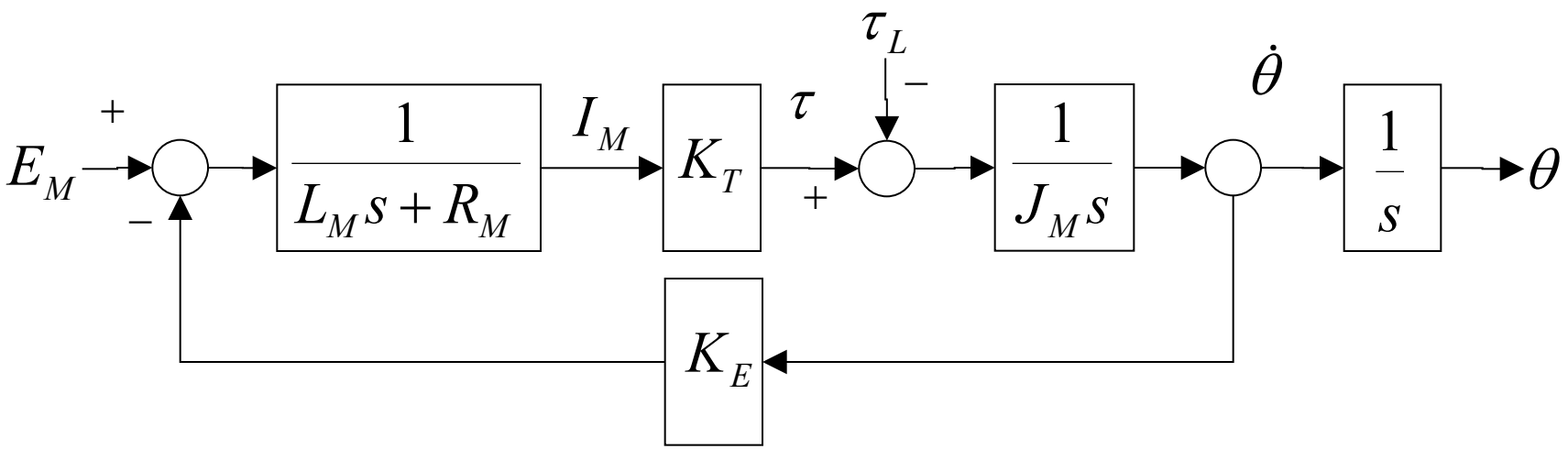
T_m

J_M

L_M

			302018	302019	302020	302021
モータ·データ（暫定仕様）						
1	定格出力	W	0.3	0.3	0.3	0.3
2	公称電圧	Volt	1.5	3.0	4.5	6.0
3	無負荷回転数	rpm	21600	21800	21800	21700
4	停動トルク	mNm	0.358	0.414	0.400	0.397
5	回転数／トルク勾配	rpm / mNm	64500	55700	57600	58100
6	無負荷電流	mA	36.3	18.3	12.2	9.12
7	起動電流	mA	576	333	215	16
8	端子間抵抗	Ohm	2.61	9.00	21.0	37.5
9	最大許容回転数	rpm	23500	23500	23500	23500
10	最大連続電流	mA	355	191	125	93
11	最大連続トルク	mNm	0.198	0.214	0.210	0.210
12	公称電圧時最大出力	W	0.196	0.230	0.222	0.220
13	最大効率	%	57	60	59	59
14	トルク定数	mNm / A	0.657	1.31	1.97	2.63
15	回転数定数	rpm / V	15400	7690	5120	3840
16	機械的時定数	ms	143	124	128	129
17	ロータ慣性モーメント	gcm^2	0.014	0.016	0.015	0.015
18	端子間インダクタンス	mH	0.02	0.07	0.16	0.29
19	熱抵抗（ハウジング／周囲間）	K / W	130	130	130	130
20	熱抵抗（ロータ／ハウジング間）	K / W	16.2	16.2	16.2	16.2
21	巻線熱時定数	s	1.3	1.5	1.4	1.4

Block diagram of DC motor
DCモータのブロック線図



CC BY

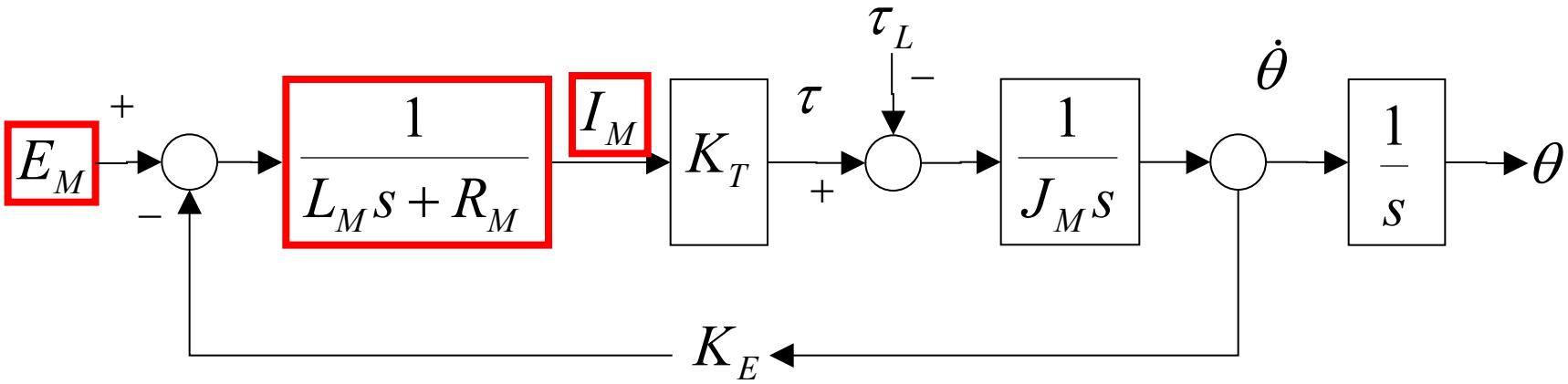
Example of Block Diagram Drawing (1/3)

$$E_M = R_M I_M + K_E \dot{\theta}(t) + L_M \frac{dI_M(t)}{dt}$$

$$E_M(s) = R_M I_M(s) + K_E \dot{\theta}(s) + L_M s I_M(s)$$

$$= (R_M + sL_M) I_M(s) + K_E \dot{\theta}(s)$$

$$E_M(s) - K_E \dot{\theta}(s) = (R_M + sL_M) I_M(s)$$



CC BY

Example of Block Diagram Drawing (2/3)

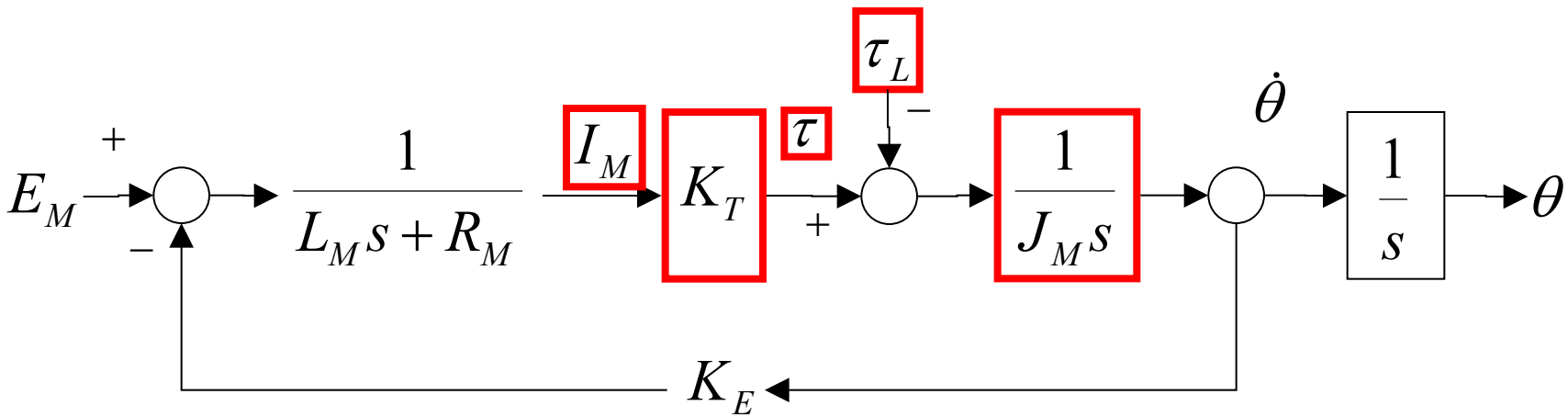
$$\tau(t) = K_T I_M(t)$$

$$\tau(s) = K_T I_M(s)$$

$$\tau(t) = J_M \ddot{\theta}(t) + \tau_L$$

$$J_M \ddot{\theta}(t) = \tau(t) - \tau_L$$

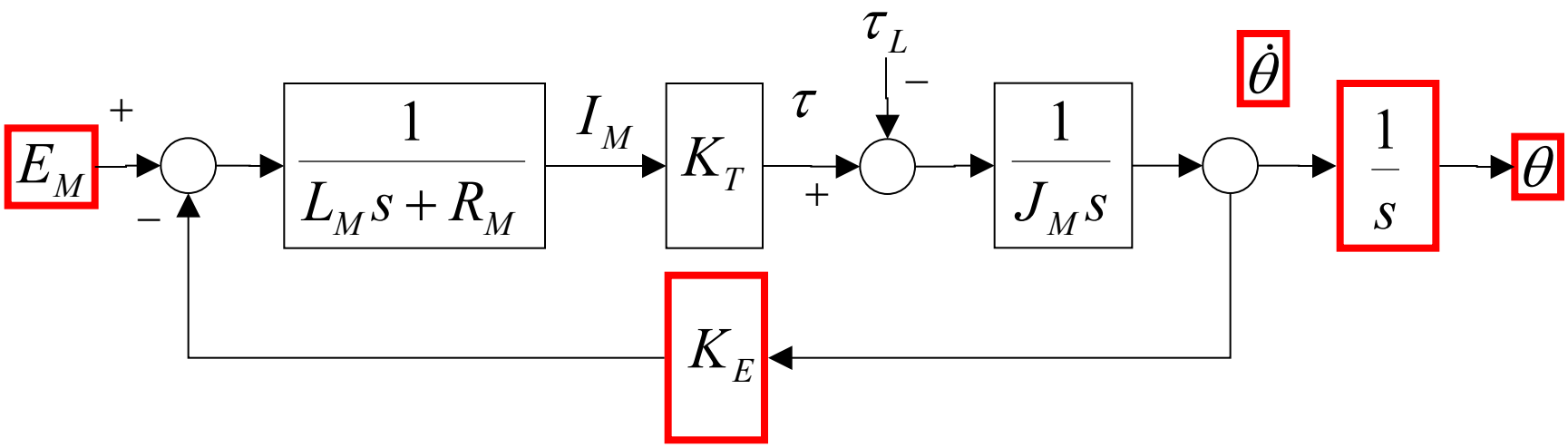
$$J_M s \dot{\theta}(s) = \tau(s) - \tau_L$$



CC BY

Example of Block Diagram Drawing (3/3)

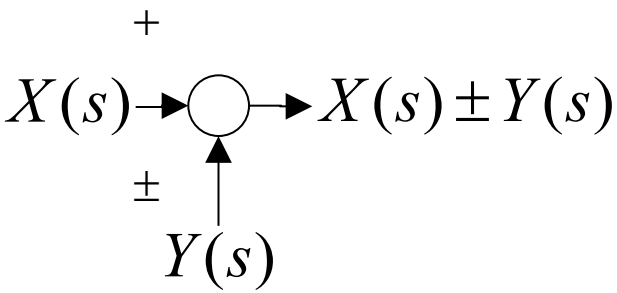
$$E_M(s) - K_E\dot{\theta}(s) = (R_M + sL_M)I_M(s) \qquad \theta(s) = \frac{1}{s}\dot{\theta}(s)$$



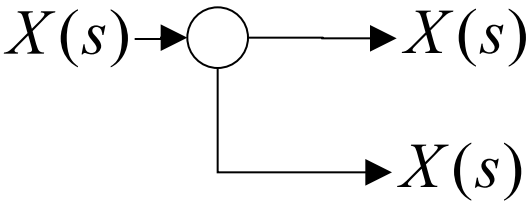
CC BY

Element of Block Diagram
ブロック線図の要素

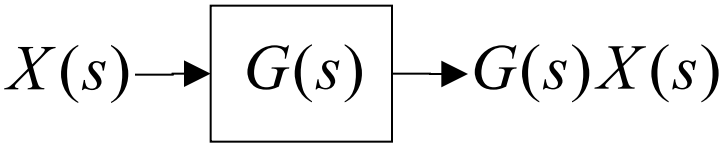
Join / 合流



Split / 分岐



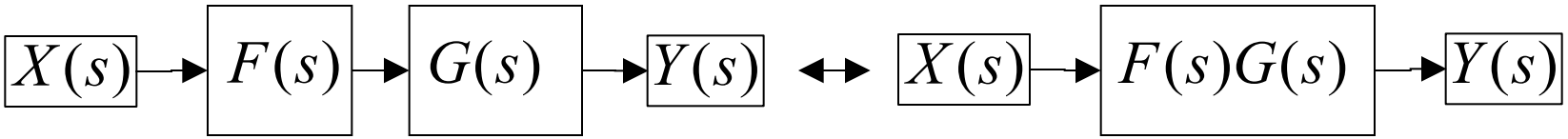
Transfer function / 伝達関数



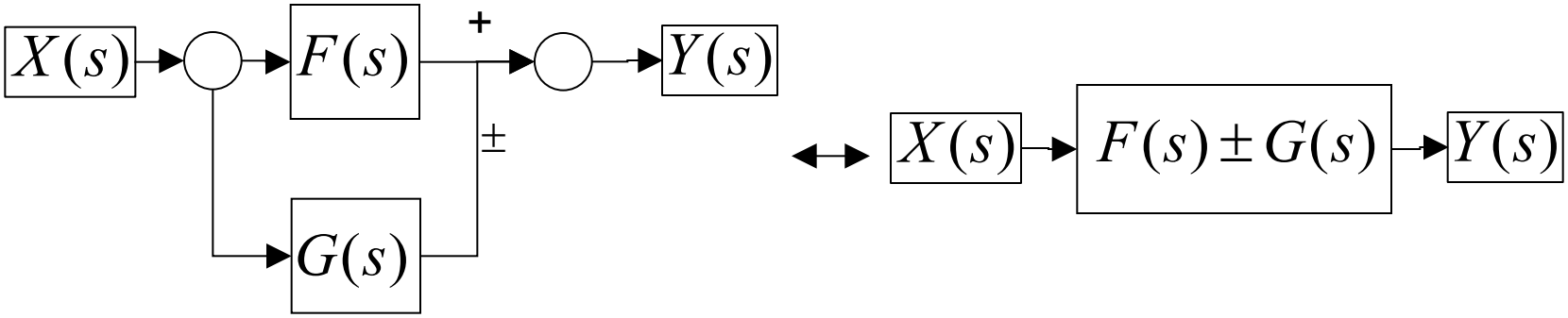
CC BY

Simplification of Block Diagram
ブロック線図の簡略化

Series connection / 直列結合



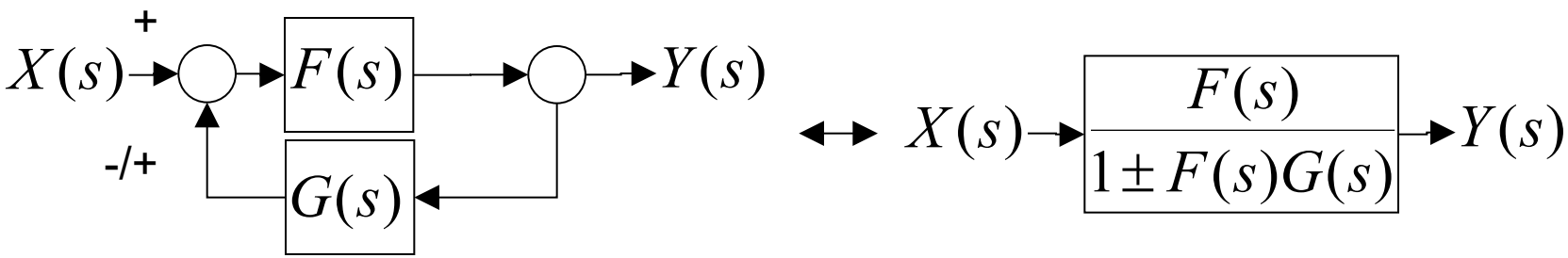
Parallel connection / 並列結合



CC BY

Simplification of Block diagram
ブロック線図の簡略化

Feedback connection / フィードバック結合



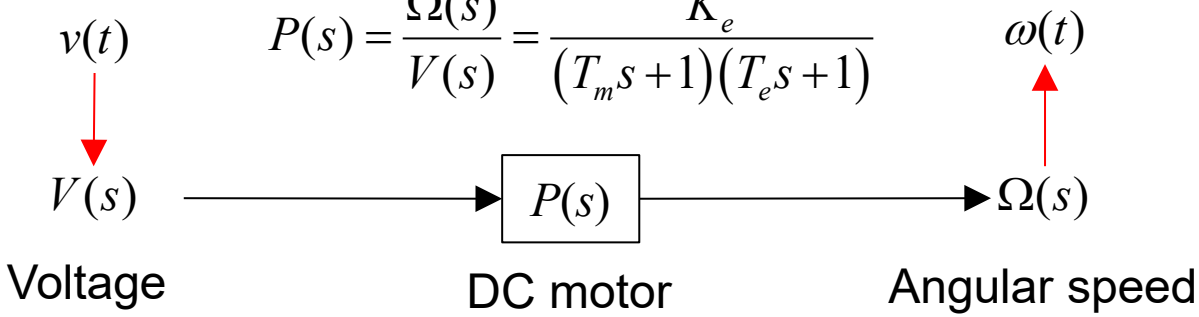
Step Response of DC Motor Transfer Function

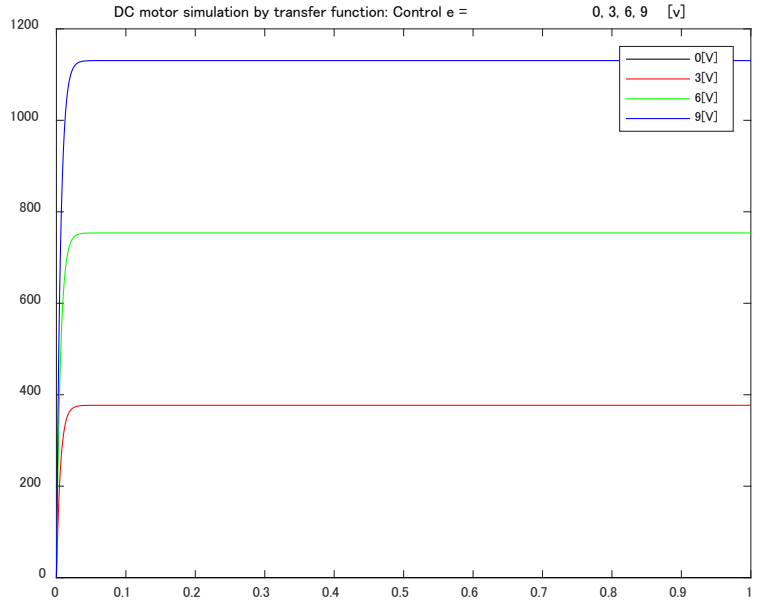
DCモータの伝達関数のステップ応答

CC BY

Time Domain Output of Transfer Function
伝達関数の時間領域出力

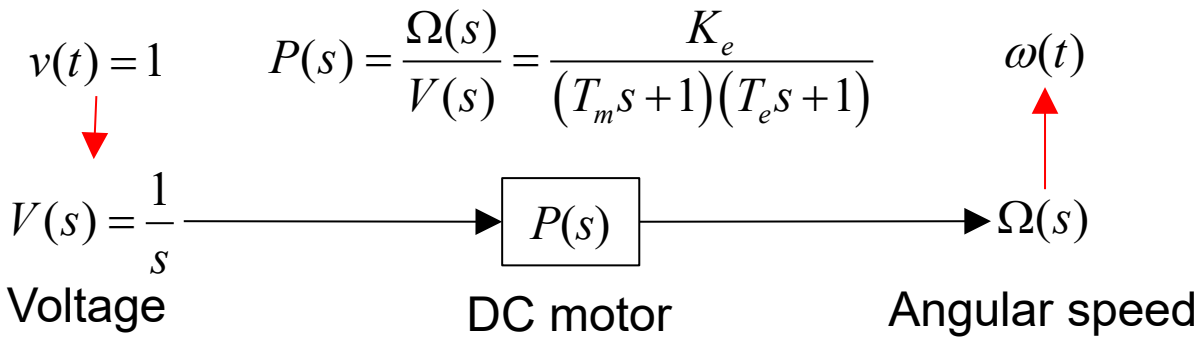
$$P(s) = \frac{\Omega(s)}{V(s)} = \frac{K_e}{(T_m s + 1)(T_e s + 1)}$$





CC BY

Step Response of Transfer Function
伝達関数のステップ応答



Step response
A time domain output $\omega(t)$ when a step input $V(s)$ = 1/s is applied.
Find it by taking inverse Laplace transform of a complex domain one $\Omega(s)$.

ステップ応答
ステップ入力$V(s)$=1/sを与えたときの時間領域での出力$\omega(t)$
複素領域での出力Ω(s)を逆ラプラス変換して，時間領域での出力$\omega(t)$を求める

$$\Omega(s)=\frac{K_e}{(T_m s+1)(T_e s+1)}V(s)=\frac{K_e}{(T_m s+1)(T_e s+1)s}$$

CC BY

Derivation of Step Response
ステップ応答の導出

If we have the following term in a transfer function, we can easily transform back to a time domain
次の項が伝達関数にあれば，簡単に逆ラプラス変換をすることができる

$$\frac{1}{s+a} \longrightarrow e^{-at} \qquad \frac{1}{Ts+1} = \frac{1}{T}\frac{1}{s+1/T} \longrightarrow \frac{1}{T}e^{-\frac{t}{T}}$$

Let us set $\Omega(s)$ as follows and find A, B, and C
$\Omega(s)$を次のようにおいてA, B, Cを求める

$$\Omega(s) = \frac{K_e}{(T_m s+1)(T_e s+1)s} = \frac{A}{T_m s+1} + \frac{B}{T_e s+1} + \frac{C}{s} \longrightarrow \omega(t) = \frac{A}{T_m}e^{-\frac{t}{T_m}} + \frac{B}{T_e}e^{-\frac{t}{T_e}} + C$$

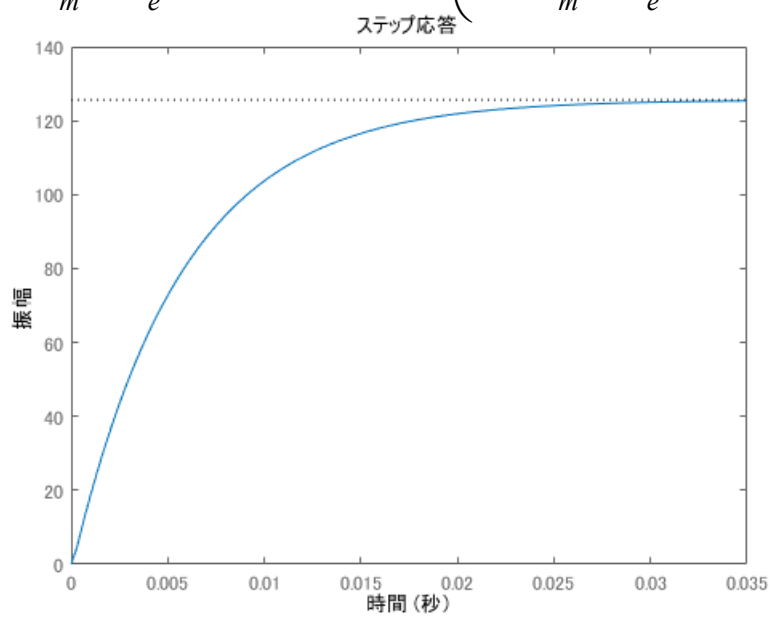
$$\frac{K_e}{(T_m s+1)(T_e s+1)s} = \frac{As(T_e s+1) + Bs(T_m s+1) + C(T_m s+1)(T_e s+1)}{(T_m s+1)(T_e s+1)s}$$
$$= \frac{(AT_e + BT_m + CT_m T_e)s^2 + (A + B + CT_m + CT_e)s + C}{(T_m s+1)(T_e s+1)s}$$

Derivation of Step Response
ステップ応答の導出

$$AT_e + BT_m + CT_mT_e = 0, A + B + CT_m + CT_e = 0, C = K_e$$

$$A = -\frac{K_eT_m^2}{T_m - T_e}, B = \frac{K_eT_e^2}{T_m - T_e}, C = K_e$$

$$\omega(t) = -\frac{K_eT_m}{T_m - T_e}e^{-\frac{t}{T_m}} + \frac{K_eT_e}{T_m - T_e}e^{-\frac{t}{T_e}} + K_e = K_e\left(1 - \frac{T_m}{T_m - T_e}e^{-\frac{t}{T_m}} + \frac{T_e}{T_m - T_e}e^{-\frac{t}{T_e}}\right)$$



$$\lim_{t\to\infty}\omega(t) = K_e$$

$$K_e = 125$$

CC BY

Summary of Step Response of DC Motor Transfer Function
DCモータの伝達関数のステップ応答のまとめ

- Angular velocity of DC motor converges to Ke exponentially
  DCモータは回転速度は，Keの値に指数関数的に収束する

- If we know all the mechanical and electrical parameters, and no external load is applied to a motor, we can predict an angular velocity for an applied voltage (Feedforward control)
  モータの機械的・電気的パラメータが既知で，無負荷のときは印加電圧と回転速度の関係が分かる（フィードフォワード制御）

- How we should control it if we do not > Introduction of feedback control
  それらが未知の場合は？フィードバック制御の導入

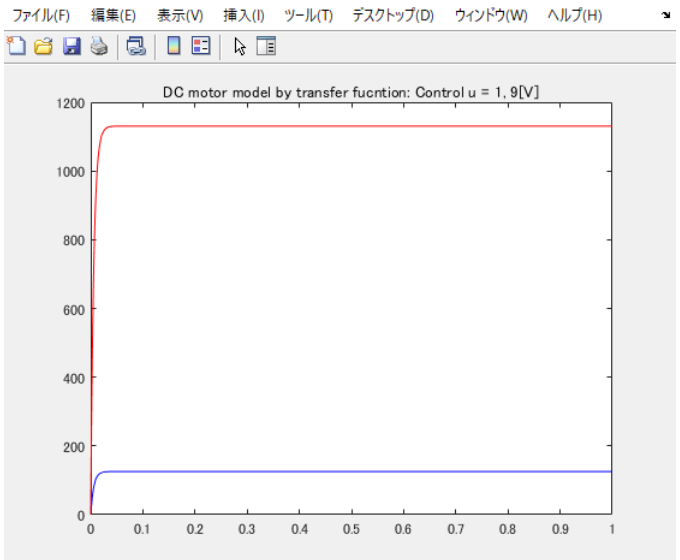
Proportional (or P) control

比例制御またはP制御

CC BY

How to Control Plant to Reference
どのように対象の出力を目標値に制御するか

Transfer function of DC motor / DCモータの伝達関数

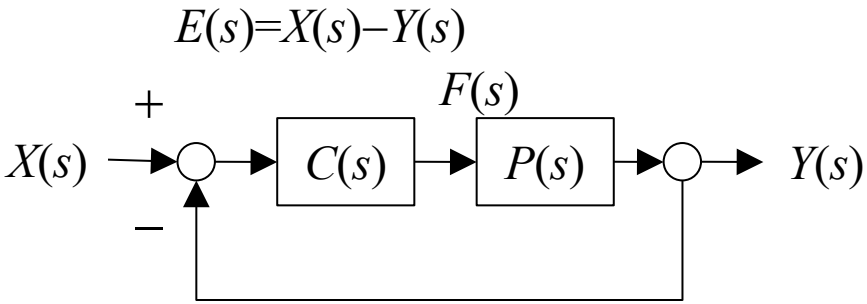


$$E_M(s) = \omega(s)K_e\left(T_mT_es^2 + T_ms + 1\right)$$
$$\approx \omega(s)K_e\left(T_ms+1\right)\left(T_es+1\right)$$

Even if we know motor parameters, it is still difficult to find a control for a reference /
モータのパラメータが分かっていても，出力が目標値になるような制御量を見つけるのは困難

We solve it by a feedback control / フィードバック制御で解決する

Feedback Control System
フィードバック制御系



- X(s): Reference value /目標値
- Y(s): Actual value/ 現在値
- E(s): Feedback amount / フィードバック量
- C(s): Controller / 制御器
- F(s): Control amount / 制御量（制御力）
- P(s): Plant / 制御対象

$$Y(s) = P(s)C(s)\big(X(s) - Y(s)\big)$$

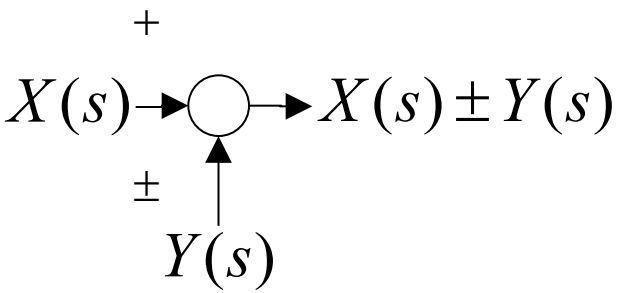
$$\big(1 + P(s)C(s)\big)Y(s) = P(s)C(s)X(s)$$

$$Y(s) = \frac{P(s)C(s)}{1 + P(s)C(s)} X(s)$$

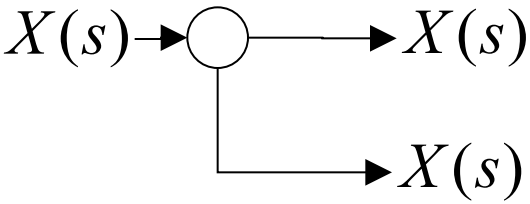
CC BY

Element of Block Diagram
ブロック線図の要素

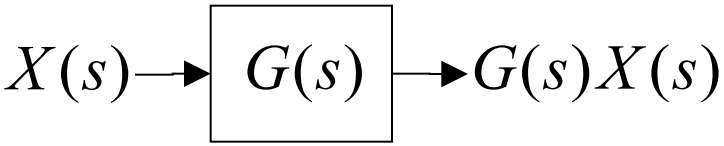
Join / 合流



Split / 分岐



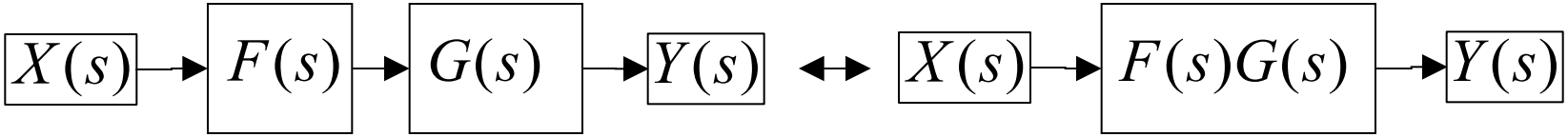
Transfer function / 伝達関数



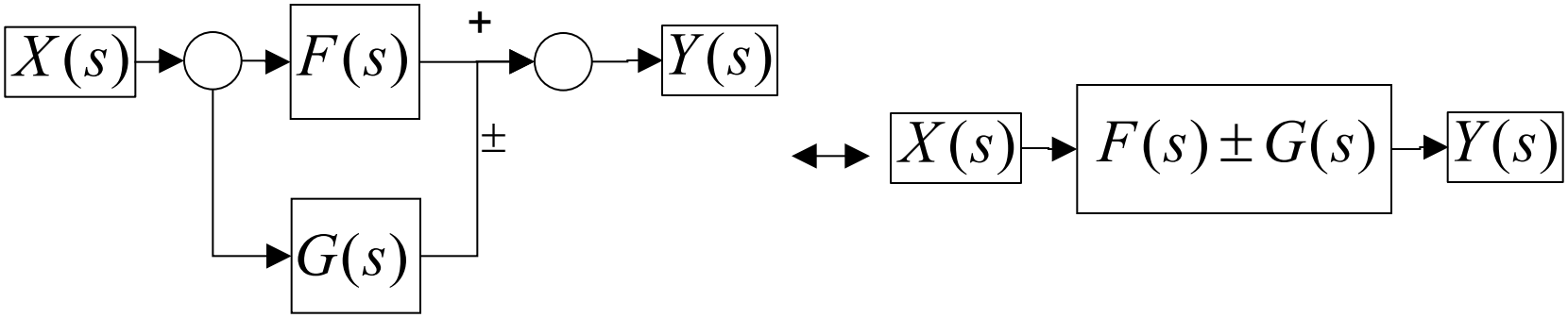
CC BY

Simplification of Block Diagram
ブロック線図の簡略化

Series connection / 直列結合



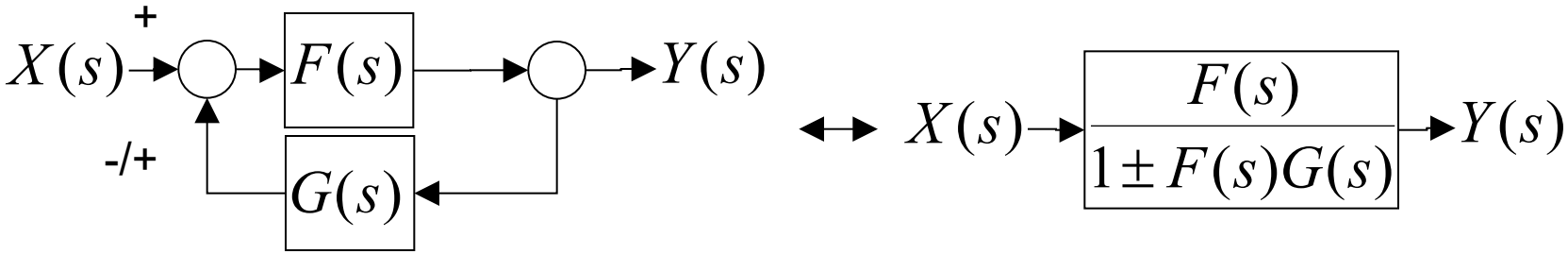
Parallel connection / 並列結合



CC BY

Simplification of Block Diagram
ブロック線図の簡略化

Feedback connection / フィードバック結合



Matlab Code for P-control of DC Motor

DCモータのP制御のMatlabコード

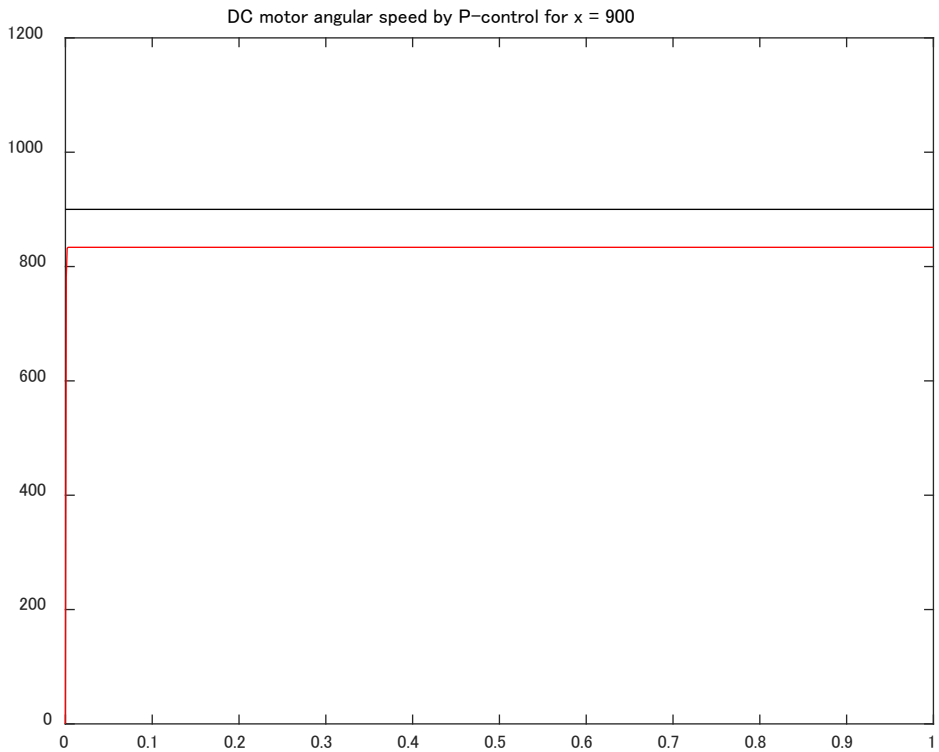
```
% Motor parameters
R = 0.309;
L = 0.0275 * 1E-3;
Tk = 5.68 * 1E-3;
Te = L/R;
Ke = 125.6;

% Set it as transfer function
P = tf([Ke], [Te*Tk, Tk+Te, 1]);

% Simulation setting
t = [0:0.001:1];
```

Matlab Code for P-control of DC Motor

DCモータのP制御のMatlabコード

```
% Set a P-controller
C = 0.1;

% Set a reference
X = 900 * ones(size(t));

% Whole system
g = feedback(C*P, 1);

% Simulation
Y = lsim(g, X, t);

% Plot
figure(1);
plot(t, Y);
```

Results of Matlab Simulation
Matlabシミュレーションの結果

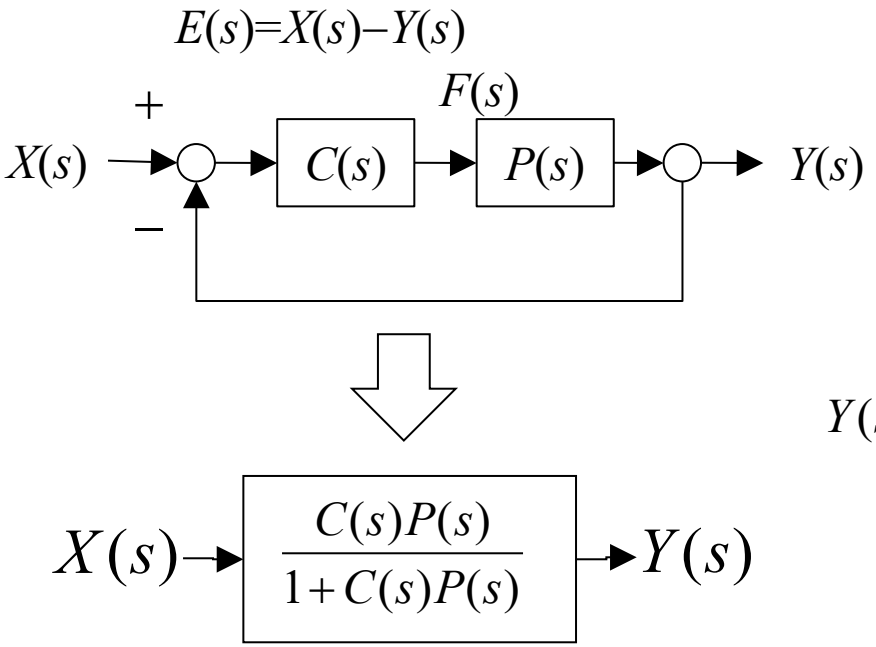


It does not converge to the reference of 900. We will discuss it later.
参照（目標）の900に収束していない．これについては後で議論する

CC BY

Step Response of P-Control System

P制御系のステップ応答



$$x(t)=\begin{cases}1, & t\geq 0,\\ 0, & \text{otherwise},\end{cases}\quad X(s)=\frac{1}{s},$$

$$C(s)=k,\ P(s)=\frac{K_e}{(T_m s+1)(T_e s+1)}$$

Suppose we set a reference (target) as a step function. Find a time domain output $y(t)$ by taking inverse Laplace transform of $Y(s)$

参照がステップ関数とする．このときの出力$Y(s)$の逆ラプラス変換をとることにより，時間領域での出力$y(t)$を求める

$$Y(s)=\frac{C(s)P(s)}{1+C(s)P(s)}X(s)=\frac{k\dfrac{K_e}{(T_m s+1)(T_e s+1)}}{1+k\dfrac{K_e}{(T_m s+1)(T_e s+1)}}\frac{1}{s}$$

$$=\frac{kK_e}{(T_m s+1)(T_e s+1)+kK_e}\frac{1}{s}$$

$$=\frac{kK_e}{T_m T_e s^2+(T_m+T_e)s+(1+kK_e)}\frac{1}{s}$$

$$=\frac{(As+B)kK_e}{T_m T_e s^2+(T_m+T_e)s+(1+kK_e)}+\frac{C}{s}$$

CC BY

Inverse Laplace Transform
逆ラプラス変換

$$\frac{s+\alpha}{(s+\alpha)^2+\omega^2} \longrightarrow e^{-\alpha t}\cos\omega t \qquad \frac{1}{s} \longrightarrow \begin{cases} 1, & t \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

$$Y(s) = \frac{(As+B)kK_e}{T_mT_es^2+(T_m+T_e)s+(1+kK_e)} + \frac{C}{s} = \frac{AkK_es^2+BkK_es+C\left(T_mT_es^2+(T_m+T_e)s+(1+kK_e)\right)}{\left(T_mT_es^2+(T_m+T_e)s+(1+kK_e)\right)s}$$

$$= \frac{(AkK_e+CT_mT_e)s^2+\left(BkK_e+C(T_m+T_e)\right)s+C(1+kK_e)}{\left(T_mT_es^2+(T_m+T_e)s+(1+kK_e)\right)s} = \frac{kK_e}{\left(T_mT_es^2+(T_m+T_e)s+(1+kK_e)\right)s}$$

$$\begin{cases} AkK_e+CT_mT_e=0 \\ C(T_m+T_e)+BkK_e=0 \\ C(1+kK_e)=kK_e \end{cases} \longrightarrow \begin{cases} A=-\dfrac{T_mT_e}{1+kK_e} \\ B=-\dfrac{T_m+T_e}{1+kK_e} \\ C=\dfrac{kK_e}{1+kK_e} \end{cases}$$

CC BY

Inverse Laplace Transform
逆ラプラス変換

$$Y(s)=\frac{kK_e}{1+kK_e}\left(\frac{1}{s}-\frac{T_mT_es+(T_m+T_e)}{T_mT_es^2+(T_m+T_e)s+(1+kK_e)}\right)$$

$$=\frac{kK_e}{1+kK_e}\left(\frac{1}{s}-\frac{s+\dfrac{T_m+T_e}{T_mT_e}}{s^2+\left(\dfrac{T_m+T_e}{T_mT_e}\right)s+\left(\dfrac{1+kK_e}{T_mT_e}\right)}\right)$$

$$=\frac{kK_e}{1+kK_e}\left(\frac{1}{s}-\frac{s+\dfrac{T_m+T_e}{T_mT_e}}{\left(s+\dfrac{T_m+T_e}{2T_mT_e}\right)^2+\left(\dfrac{1+kK_e}{T_mT_e}\right)-\left(\dfrac{T_m+T_e}{2T_mT_e}\right)^2}\right)$$

$$y(t)=\frac{kK_e}{1+kK_e}\left(1-e^{-\frac{T_m+T_e}{T_mT_e}t}\cos\left(\sqrt{\left(\frac{1+kK_e}{T_mT_e}\right)-\left(\frac{T_m+T_e}{2T_mT_e}\right)^2}\,t\right)\right)$$

$$\frac{s+\alpha}{(s+\alpha)^2+\omega^2}$$

$$\downarrow$$

$$e^{-\alpha t}\cos\omega t$$

$$\frac{1}{s}$$

$$\downarrow$$

$$\begin{cases}1, & t\geq 0,\\ 0, & \text{otherwise.}\end{cases}$$

CC BY

Analysis of Step Response of P-Control
P制御のステップ応答の解析

$$y(t)=\frac{kK_e}{1+kK_e}\left(1-e^{-\frac{T_m+T_e}{T_mT_e}t}\cos\left(\sqrt{\left(\frac{1+kK_e}{T_mT_e}\right)-\left(\frac{T_m+T_e}{2T_mT_e}\right)^2}\,t\right)\right)$$

Steady State Error: An error between a reference and actual output as time goes to infinity.

定常誤差：時間が無限大になったときの参照と実際の出力の差

$$e(t)=x(t)-y(t),\lim_{t\to\infty}e(t)=\lim_{t\to\infty}\left(1-\frac{kK_e}{1+kK_e}\right)=\frac{1}{1+kK_e}$$

We always have the steady state error in P-control.

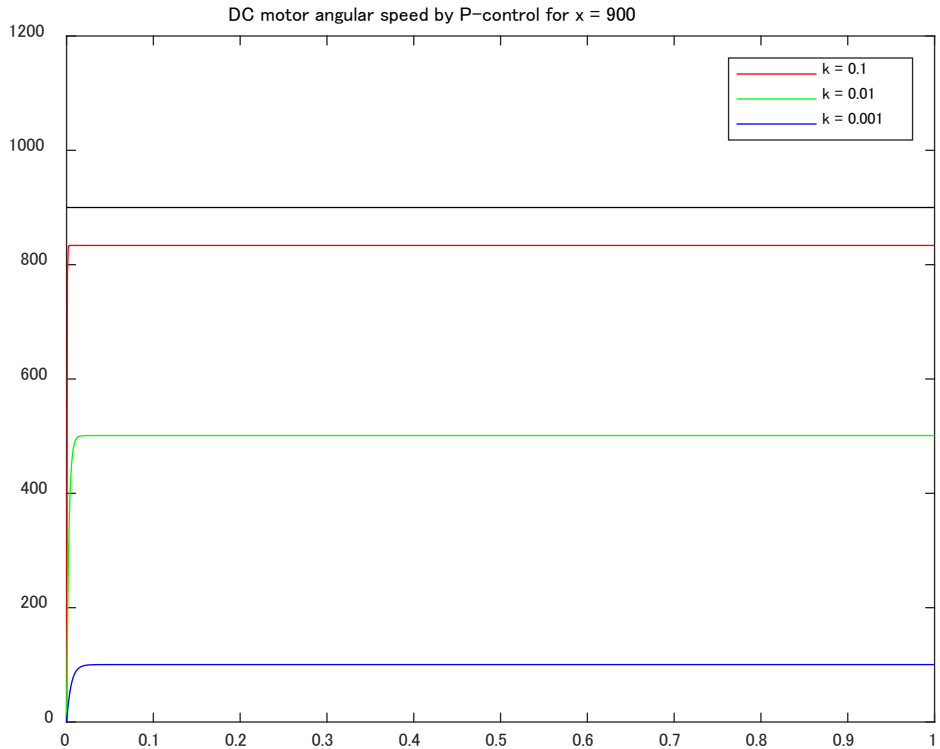
If we increase a gain k, it decreases, but it never goes to 0.

P制御では必ず定常誤差が残る.

比例制御のゲインkを増加すると，定常誤差は減少するがゼロにはならない

$$k=1\to\frac{1}{1+K_e},k=10\to\frac{1}{1+10K_e},k=100\to\frac{1}{1+100K_e},\cdots$$

Steady State Error by Different Gain k
異なるゲインkによる定常誤差



CC BY

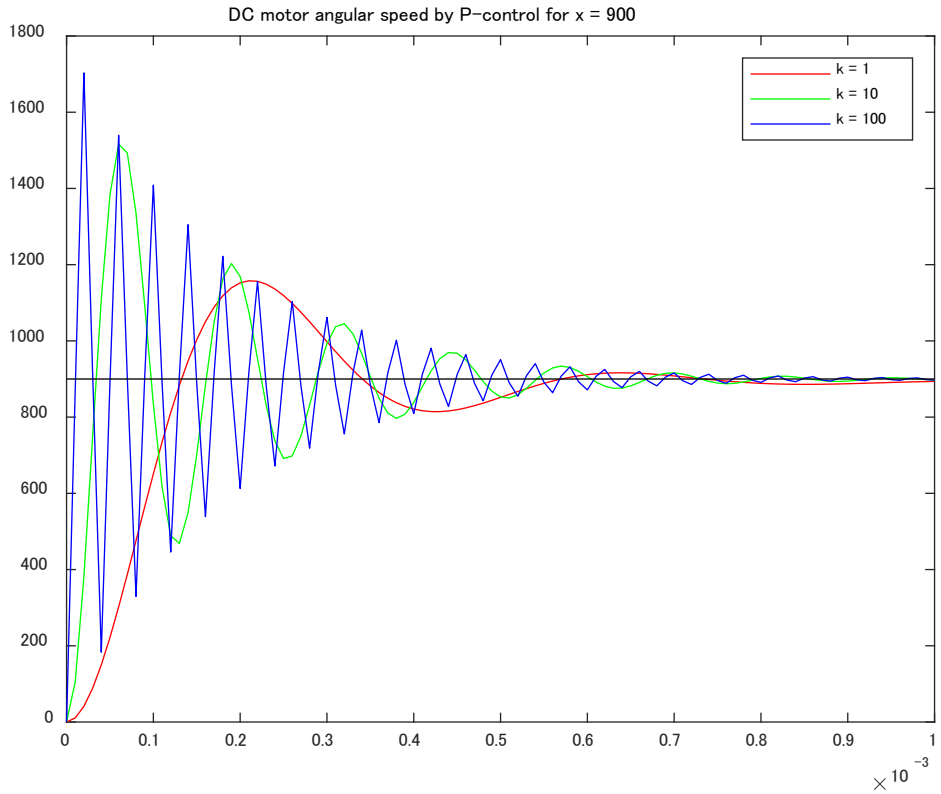
Analysis of Step Response of P-Control

P制御のステップ応答の解析

$$y(t) = \frac{kK_e}{1+kK_e}\left(1 - e^{-\frac{T_m+T_e}{T_mT_e}t}\cos\left(\sqrt{\left(\frac{1+kK_e}{T_mT_e}\right) - \left(\frac{T_m+T_e}{2T_mT_e}\right)^2}\,t\right)\right)$$

If we increase a gain k, we have a high oscillation frequency w, which causes an unstable control system.
振動: ゲイン*k*を増加すると，応答の振動性が増大し（周波数が速くなり），制御が不安定になる

Oscillation Frequency by Different k
異なるゲインkによる振動周波数



CC BY

Class of Plants (Control Targets)
制御対象のクラス

- First order lag: Transfer function has the following form (a first order differential eqn.w.r.t. time)
  1次遅れ：伝達関数が次の形式を持つ（時間に関する一階微分方程式）

$$F(s) = \frac{1}{s+a} \, or \, \frac{1}{Ts+1}$$

- Second order lag: Transfer function has the following form (a second order differential eqn.w.r.t. time)
  2次遅れ：伝達関数が次の形式を持つ（時間に関する二階微分方程式）

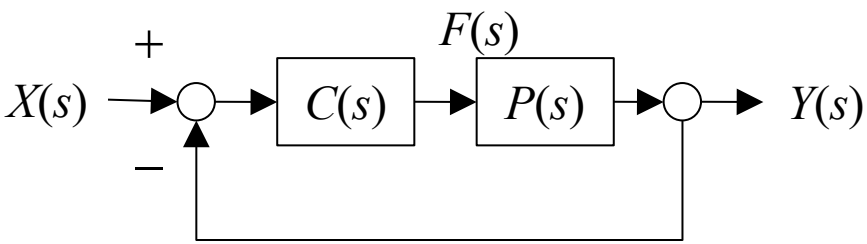
$$F(s) = \frac{1}{s^2+as+b} \, or \, \frac{1}{(T_1 s+1)(T_2 s+1)}$$

CC BY

Steady Sate Error in S Domain

S（複素数）領域での定常誤差の計算

$E(s)=X(s)-Y(s)$



$$E(s) = X(s) - Y(s)$$
$$= X(s) - P(s)C(s)E(s)$$
$$E(s) = \frac{1}{1+P(s)C(s)}X(s)$$

$$P(s) = \frac{1}{as^2+bs+1},\ C(s)=k,\ X(s)=\frac{1}{s}$$
$$E(s) = \frac{1}{1+P(s)C(s)}X(s) = \frac{1}{1+\dfrac{k}{as^2+bs+1}}\frac{1}{s}$$
$$= \frac{as^2+bs+1}{as^2+bs+1+k}\frac{1}{s}$$
$$\lim_{s\to 0} sE(s) = \lim_{s\to 0} s\frac{as^2+bs+1}{as^2+bs+1+k}\frac{1}{s} = \frac{1}{1+k}$$

Final value theorem
最終値の定理

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s)$$

CC BY

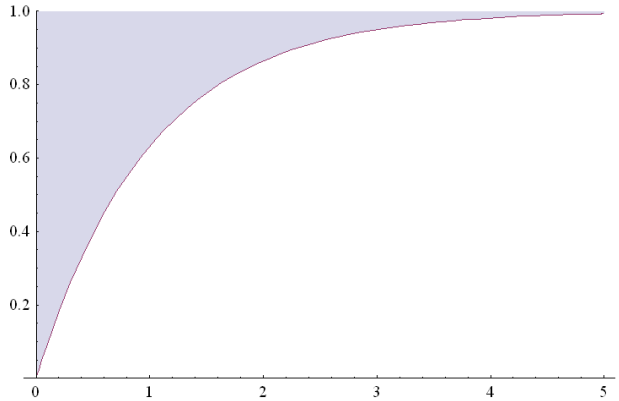
Summary of P-Control
P制御のまとめ

- We have studies P-control as the simplest feedback control system.
  最も単純なフィードバック制御システムとしてP制御を学んだ
  - It can control an output close to a reference even if we do not know an exact system model
    正確なシステムのモデルがなくても出力を参照に近づけることができる
  - However, we always have a steady state error by P-Control
    しかし，P制御では常に定常誤差が発生する
  - We have learned it by analysis of transfer functions and simulation by matlab
    伝達関数の解析とmatlabによるシミュレーションでそれを学んだ

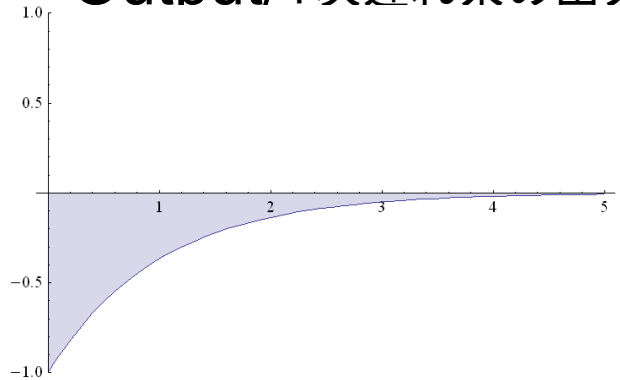
Integral (or I) control
積分制御またはI制御

CC BY

Problem in P-control
P制御の問題



Output/1次遅れ系の出力



Cumulation of error(Integral)/ 目標値との誤差の累積（積分）

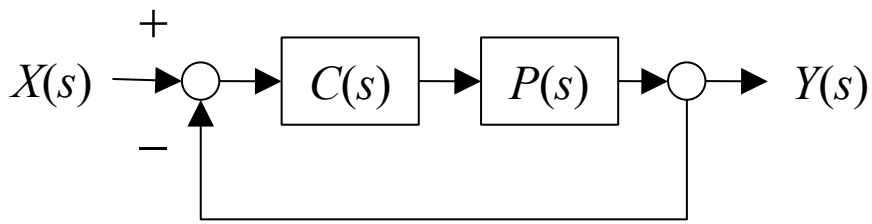
- P control always remains a steady state error
  P制御では常に定常誤差が残ってしまう
- Consider cumulative error and minimize it
  制御開始時から現時点までの目標量の積分（面積を）考え，これを最小化する

CC BY

Integral control

I（積分）制御

$$E(s)=X(s)-Y(s)$$



$$P(s)=\frac{K_e}{T_mT_es^2+\left(T_m+T_e\right)s+1},\ C(s)=\frac{k_i}{s},\ X(s)=\frac{1}{s}$$

$$E(s)=\frac{1}{1+P(s)C(s)}X(s)=\frac{1}{1+\dfrac{K_e}{T_mT_es^2+\left(T_m+T_e\right)s+1}\dfrac{k_i}{s}}\frac{1}{s}=\frac{s}{s+\dfrac{K_ek_i}{T_mT_es^2+\left(T_m+T_e\right)s+1}}\frac{1}{s}$$

$$\lim_{s\to 0}sE(s)=\lim_{s\to 0}s\frac{s}{s+\dfrac{K_ek_i}{T_mT_es^2+\left(T_m+T_e\right)s+1}}\frac{1}{s}=\lim_{s\to 0}\frac{s}{s+\dfrac{K_ek_i}{T_mT_es^2+\left(T_m+T_e\right)s+1}}=\frac{0}{0+K_ek_i}=0$$

We can reduce a steady state error to zero by I-control.
I制御によって定常誤差を0にすることができる

CC BY

Step Response of PI-Control of DC Motor

2次遅れ要素のPI制御

$$P(s)=\frac{K_e}{T_mT_es^2+(T_m+T_e)s+1},\ C(s)=\left(k_p+\frac{k_i}{s}\right),\ X(s)=\frac{1}{s}$$

$$Y(s)=\frac{P(s)C(s)}{1+P(s)C(s)}X(s)=\frac{\left(k_p+\frac{k_i}{s}\right)\frac{K_e}{T_mT_es^2+(T_m+T_e)s+1}}{1+\left(k_p+\frac{k_i}{s}\right)\frac{K_e}{T_mT_es^2+(T_m+T_e)s+1}}\frac{1}{s}\longrightarrow y(t)$$

$$=\frac{K_e\left(k_p+\frac{k_i}{s}\right)}{T_mT_es^2+(T_m+T_e)s+1+\left(k_p+\frac{k_i}{s}\right)}\frac{1}{s}$$

Because complicated, it is almost impossible to represent a closed form an output function in a time domain y(t). We evaluate it by a steady state error and numerical simulation

複雑なので時間領域で出力関数y(t)を閉じた形で表現するのは不可能→定常誤差と数値計算により評価する

CC BY

Summary of I-Control and PI-Control
I制御とPI制御のまとめ

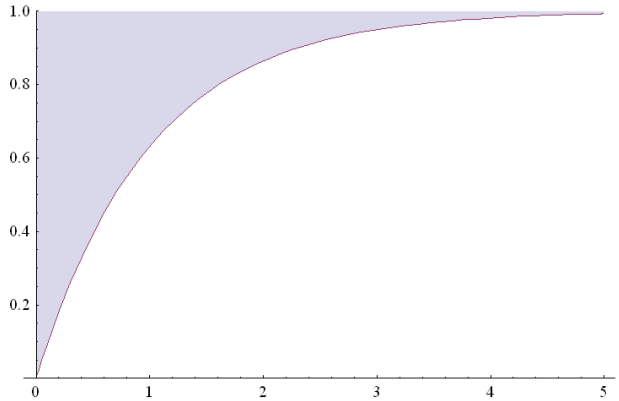
- We have studies I-control and PI-control as an improved feedback control system.
  改善されたフィードバック制御システムとしてI制御とPI制御を学んだ
  - It can converge to a reference.
    出力を参照に一致することができる
  - However, it converges slowly
    しかし，収束速度が遅い

Differential (or D) control

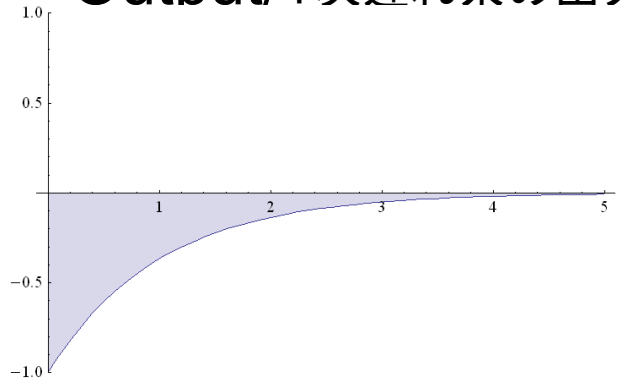
微分制御またはD制御

CC BY

Problem in PI-control
PI制御の問題



Output/1次遅れ系の出力



Cumulation of error(Integral)/ 目標値との誤差の累積（積分）

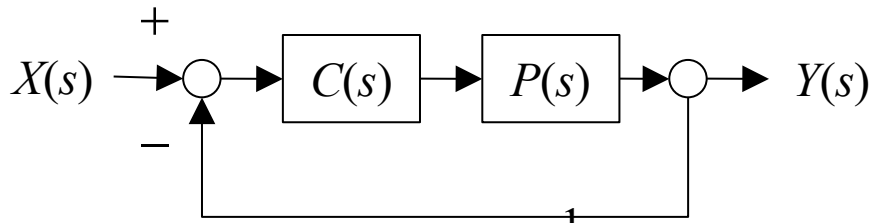
- PI-control converges to a reference, but it is slow convergence
  PI制御では定常誤差に収束するが，収束速度が遅い
- To speed it up, we consider differential element
  収束を速くするために微分要素を考える

CC BY

Differential Control

D（微分）制御

$$E(s)=X(s)-Y(s)$$



$$P(s)=\frac{K_e}{T_mT_es^2+\left(T_m+T_e\right)s+1},\ C(s)=k_ds,\ X(s)=\frac{1}{s}$$

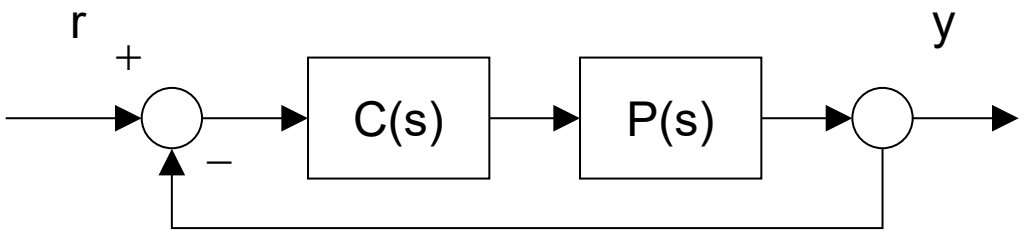
$$E(s)=\frac{1}{1+P(s)C(s)}X(s)=\frac{1}{1+\dfrac{K_ek_ds}{T_mT_es^2+\left(T_m+T_e\right)s+1}}\frac{1}{s}=\frac{T_mT_es^2+\left(T_m+T_e\right)s+1}{T_mT_es^2+\left(T_m+T_e\right)s+1+K_ek_ds}\frac{1}{s}$$

$$\lim_{s\to 0}sE(s)=\lim_{s\to 0}s\frac{T_mT_es^2+\left(T_m+T_e\right)s+1}{T_mT_es^2+\left(T_m+T_e\right)s+1+K_ek_ds}\frac{1}{s}=\lim_{s\to 0}\frac{1}{1+K_ek_ds}=\frac{1}{1}=1$$

D-only-control does not improve a steady state error. Actually, it diverges
D制御単体では定常誤差は改善しない．実際には発散する.

PD Control
PD制御

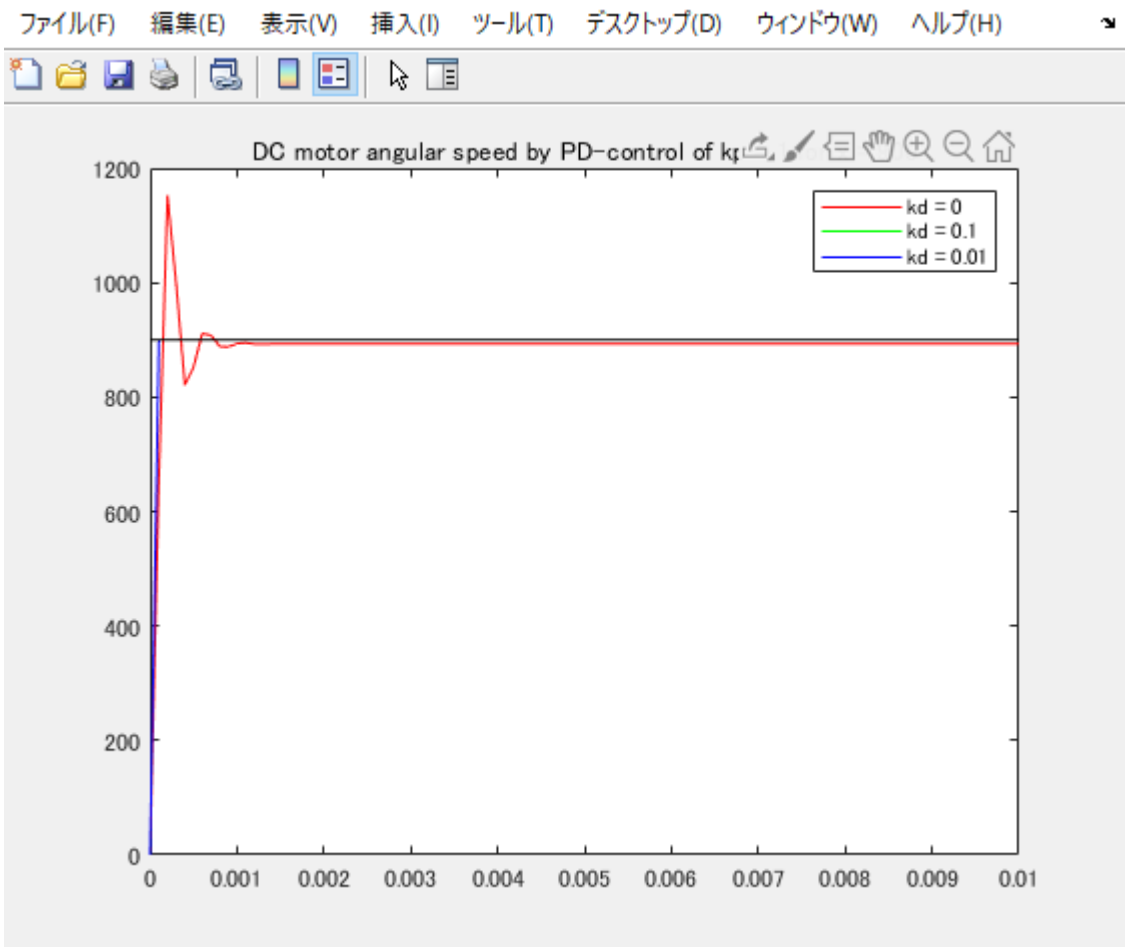
- Control by P and D component to control / 制御に比例要素Pと微分要素Dによる制御
- Steady state error is same as P-Control/定常誤差はP制御と同じ
- But it converges much faster (or peaky, unstable if we put a wrong differential gain)/しかし収束速度が改善される（ときとして不安定）



$$C(s) = k_p + k_d s$$

Matlab Example of PD Control

PD制御のMatlabの例




```
% Simulation setting
t = [0 : 0.0001 : 0.1];

% Set a PD-controller
C1 = 1 + 0*tf([1 0], [1]);
C2 = 1 + 1*tf([1 0], [1]);;
C3 = 1 + 10*tf([1 0], [1]);;
```

CC BY

Summary of D-Control and PD-Control
D制御とPD制御のまとめ

- We have studies D-control and PD-control as a faster converging feedback control system.
  早く収束するフィードバック制御システムとしてD制御とPD制御を学んだ
  - It can converge faster, but it can be unstable by a gain
    出力を速く収束することができるが，ゲインによっては不安定なことがある
  - However, it involves a steady state error.
    しかし，定常誤差が残る

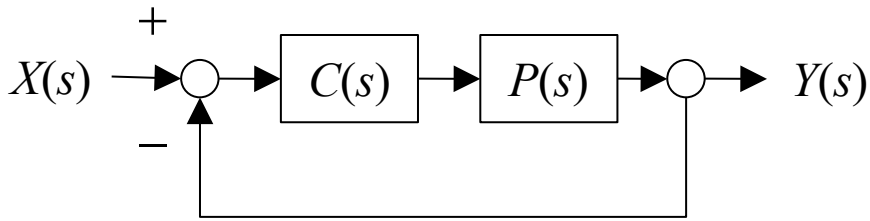
PID control

PID制御

CC BY

PID Control
PID制御

$$E(s)=X(s)-Y(s)$$



$$P(s)=\frac{K_e}{T_m T_e s^2+\left(T_m+T_e\right)s+1},\ C(s)=k_p+\frac{k_i}{s}+k_d s,\ X(s)=\frac{1}{s}$$

We can develop a good feedback controller by the combination of
以下の要素の組み合わせで良い制御器を作ることができる

- kp: a gain of P component for basic control / 基本
- ki: a gain of I component for converging to a reference (no steady state error) / 参照への収束＝定常誤差0
- kd: a gain of D component for converging faster / 速い収束

How do we tune kp, ki, and kd? どのようにゲインを見つけるか？

CC BY

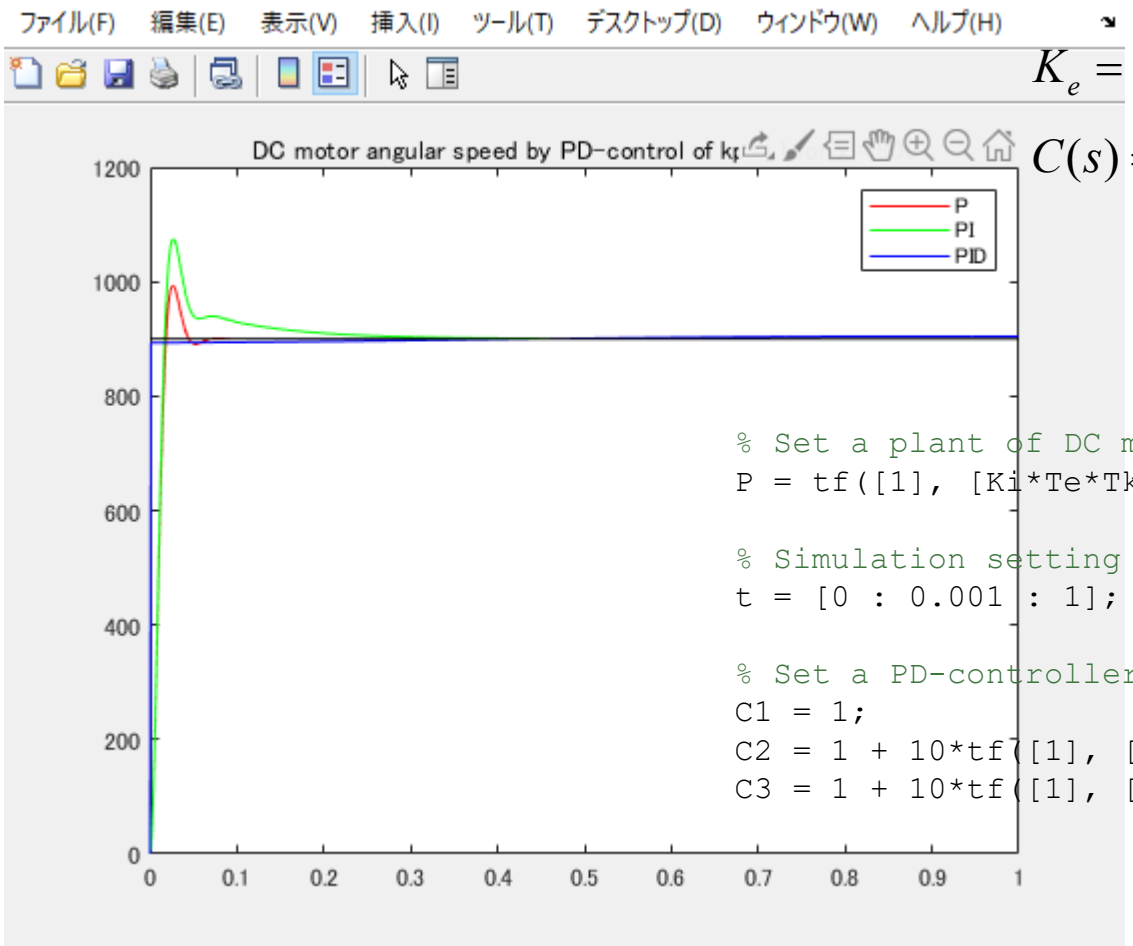
Example of DC Motor Position Control by P, PI, PID Control

DCモータの位置のP, PI, PID制御の例

$$P(s) = \frac{\Theta(s)}{V(s)} = \frac{1}{K_e s\left(T_m s + 1\right)\left(T_e s + 1\right)}$$

$$K_e = 7.96 \times 10^{-3}, T_m = 5.68 \times 10^{-3}, T_e = 8.90 \times 1$$

$$C(s) = k_p + \frac{k_i}{s} + k_d s$$




```
% Set a plant of DC motor position as transfer function
P = tf([1], [Ki*Te*Tk, Ki*(Tk+Te), Ki, 0]);

% Simulation setting
t = [0 : 0.001 : 1];

% Set a PD-controller
C1 = 1;
C2 = 1 + 10*tf([1], [1 0]);
C3 = 1 + 10*tf([1], [1 0])+ 1*tf([1 0], [1]);
```

CC BY

Empirical Way to Determine PID: Ultimate Gain method

PIDゲインの経験的な決定方法：限界感度法

- Apply a P-control and increase Kp/対象にP制御を適用し，ゲインを上げていく
- When output is pure oscillation, record it as gain K_C, and oscillation period as Tc/出力が周期的になったときのゲイン（限界ゲイン）K_cとそのときの周期（限界周期）T_cから次のようにゲインを定める
- Developed by ZieglerとNicholes

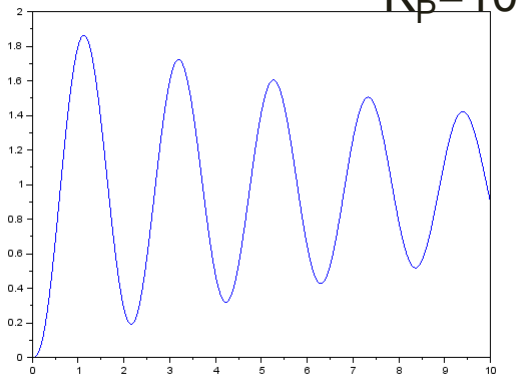
Control	**K_P**	**T_I**	**T_D**
P	$0.5K_c$		
PI	$0.45K_c$	$0.83T_c$	
PID	$0.6K_c$	$0.5T_c$	$0.125T_c$

$$C(s) = k_p + \frac{k_i}{s} + k_d s = k_p \left(1 + \frac{1}{T_i s} + T_d s \right)$$

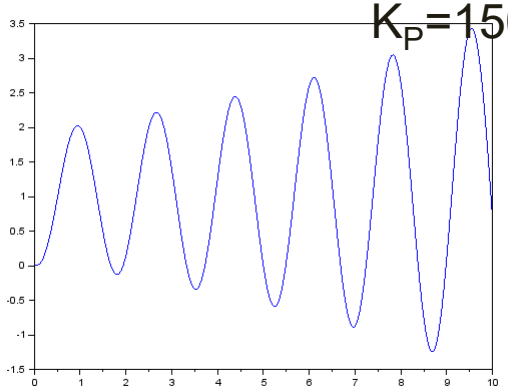
CC BY

Test many gains by trial and errors
試行錯誤的にゲインを試す

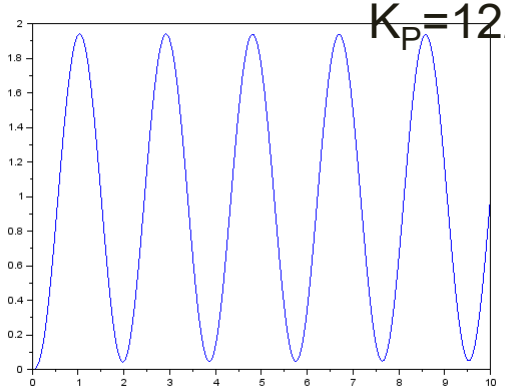
K_P=100



K_P=150



K_P=122



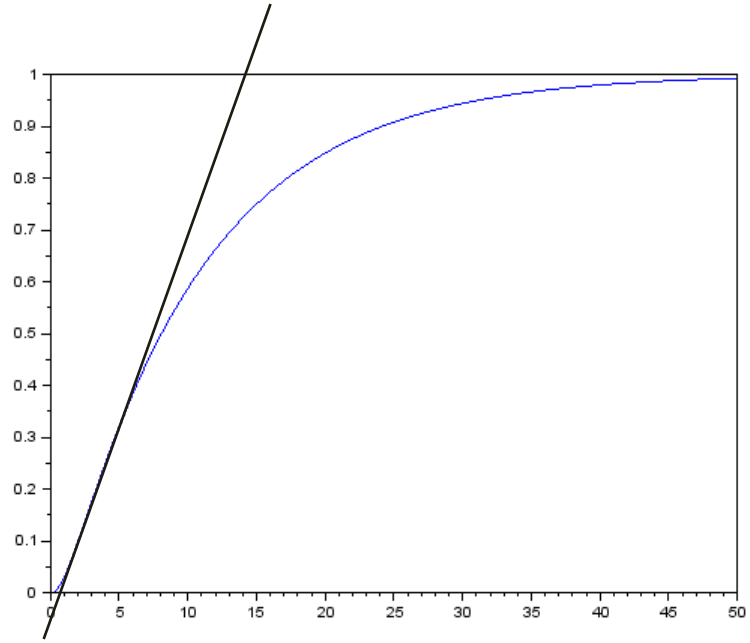
$$G(s) = \frac{1}{(0.1s+1)(s+1)(10s+1)}$$

$$K_c = 122, T_c = 1.9$$

Find the critical gain Kc and determine each of the gains following the table in the previous slide
限界感度Kcを求め，前頁の表を使いゲインを決定する

Empirical Way to Determine PID Gains: Process Reaction Curve method PIDゲインの経験的な決定方法：過渡応答法

- Apply a step input to plant with no control / 対象にステップ入力を行う（制御なし）
- Read the steepest slopeR and crossing time point L/そのときのグラフの傾きが最急部の傾きRとそれが軸と交わる時間Lを読み取る



Control	K_P	T_I	T_D
P	1/RL		
PI	0.9/RL	3.33L	
PID	1.2/RL	2L	0.5L

$$C(s) = k_p + \frac{k_i}{s} + k_d s = k_p\left(1 + \frac{1}{T_1 s} + T_d s\right)$$

R=0.08. , L=0.8
そして表からゲインを求める

CC BY

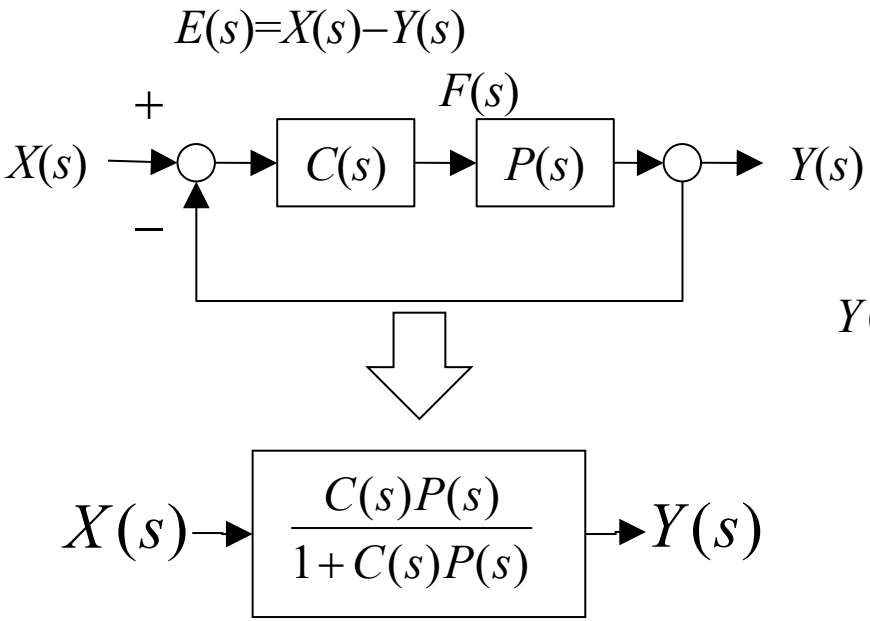
Summary of PID-Control
PID制御のまとめ

- We have studies PID-control as a feedback control system taking the advantage of
  以下の利点を持つPID制御を学んだ
  - P-control: Basic
    基本のP制御
  - I-control: Accurate (but slow)
    正確だが収束の遅いI制御
  - D-control: Fast (but unstable)
    収束が速いが不安定なD制御

CC BY

Summary of PID-Control

PID制御のまとめ



$$C(s) = k_p + \frac{k_i}{s} + k_d s = k_p\left(1 + \frac{1}{T_i s} + T_d s\right)$$

$$P(s) = \frac{K_e}{(T_m s + 1)(T_e s + 1)}$$

$$Y(s) = \frac{P(s)C(s)}{1 + P(s)C(s)} X(s)$$

$$= \frac{\dfrac{K_e}{(T_m s + 1)(T_e s + 1)} k_p\left(1 + \dfrac{1}{T_i s} + T_d s\right)}{1 + \dfrac{K_e}{(T_m s + 1)(T_e s + 1)} k_p\left(1 + \dfrac{1}{T_i s} + T_d s\right)} X(s)$$

$$= \frac{K_e k_p\left(1 + \dfrac{1}{T_i s} + T_d s\right)}{(T_m s + 1)(T_e s + 1) + K_e k_p\left(1 + \dfrac{1}{T_i s} + T_d s\right)} X(s)$$

It is too complicated to find a time-domain output of y(t) taking the inverse Laplace transform of Y(s)

Y(s)を逆ラプラス変換して，y(t)を求めるのは複雑すぎる

CC BY

Summary of PID-Control
PID制御のまとめ

- We tune a gain of P, I, D components empirically by some methods and hand tuning
  経験的にP, I, Dのゲインを調整する（手法または手調整）

Frequency Response Analysis
周波数応答の解析

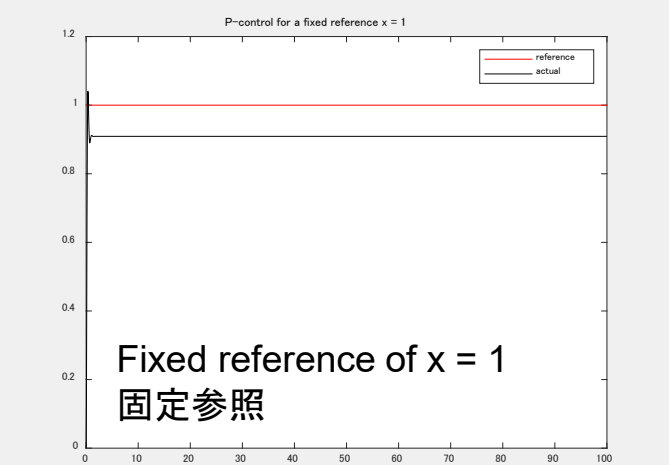
CC BY

Frequency Response Analysis
周波数応答解析

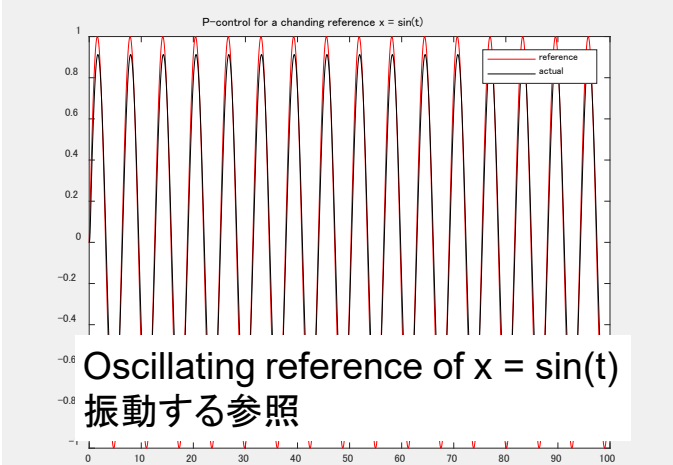
- So far, we have studied a feedback control system of a fixed reference (target).
  私たちは固定参照（目標）値でのフィードバック制御システムについて学んできた
- However, in the real situation, the reference can be changed with time. For example, if we control angular speed of wheels of a vehicle, we change the reference speed of them when the vehicle makes a turn.
  しかし制御の参照（目標）値は時間によって変わる可能性がある．例えば車両の車輪の回転速度を制御しているときは，車両が旋回するときに車輪の目標速度が変化する
- Therefore, we should analyze how the control system reacts to the changing reference
  したがって，変化する参照（目標）に対して制御システムがどのような反応するかを解析すべきである
- We model the changing speed of the reference as frequency, and we analyze the frequency response of the control system
  ここでは参照の変化速度を周波数としてモデル化し，制御システムの周波数応答を解析する

CC BY

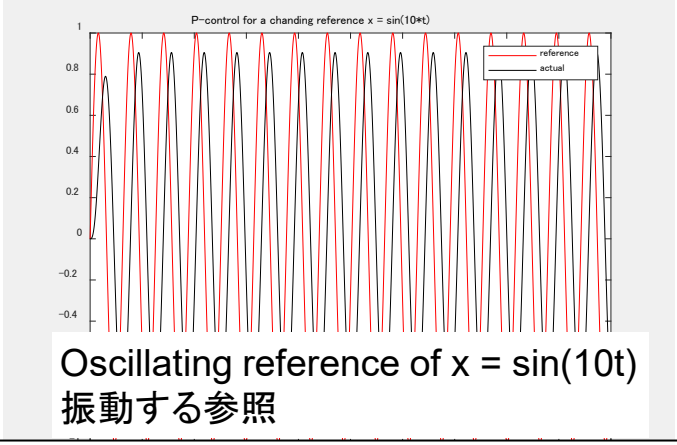
Example of Output of P-Control for Changing Reference
変化する参照に対するP制御の出力の例



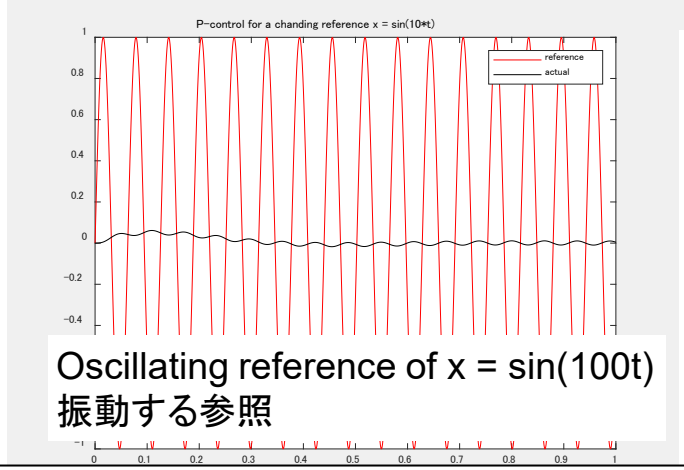
Fixed reference of x = 1
固定参照



Oscillating reference of x = sin(t)
振動する参照



Oscillating reference of x = sin(10t)
振動する参照

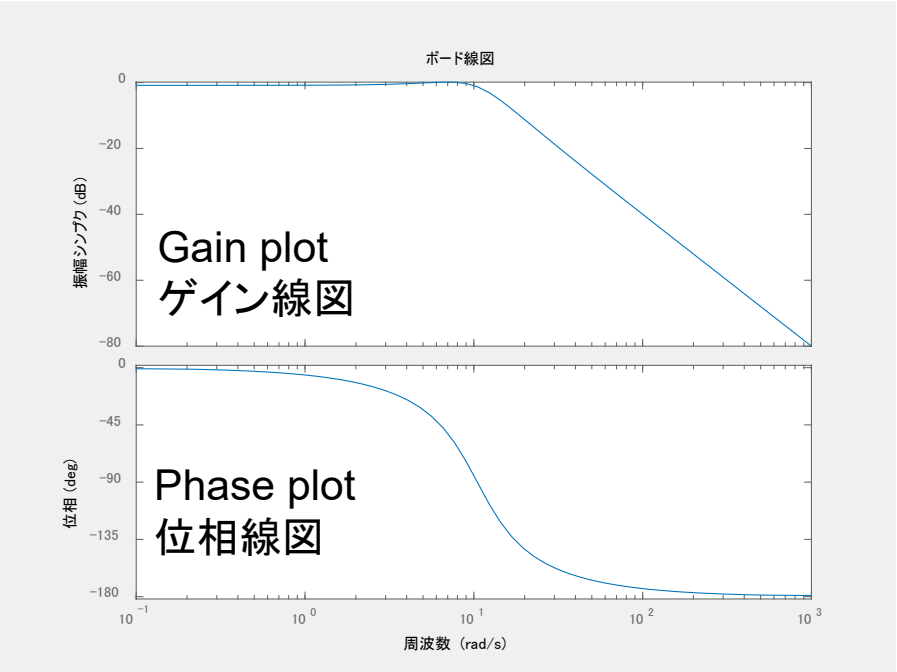


Oscillating reference of x = sin(100t)
振動する参照

As the reference changes faster (higher frequency), output gets smaller and delayed
参照の変化速度が速くなる（周波数が高くなる）につれ，出力の小さくなり，時間遅れが発生する

CC BY

Frequency Response Analysis and Bode Plot
周波数応答解析とボード線図



The vertical axis of the gain plot is DB.
The gain starts to drop at ω = 10 [rad/sec]
ゲイン線図の縦軸はデシベル
ゲインはω = 10 から減少を始める

$$20\log_{10}\frac{V_{out}}{V_{in}}[dB] = 20\log_{10}\frac{Y}{R}[dB]$$

-20dB = 10^{-1} = 0.1, -40dB = 10^{-2} = *0.01
20db = 10^{1} = 10, 40dB = 10^{2} = 100

The vertical axis of the phase plot is angle.
The phase delays as ω goes higher.
It delays about 90 deg. at ω = 10
位相線図の縦軸は角度
位相は周波数が高くなるにつれ徐々に遅れ
ω = 10 では約90度遅れる

The Bode plot shows the gain and phase of the output for the frequency, which represents the changing speed of the input(reference).
ボード線図は入力（この場合は参照）に周波数に対する，出力のゲイン（比率）と位相（時間をくれを表す）

CC BY

Principle of Bode Plot
ボード線図の原理

P(s) : Transfer function
s = τ + jω:
- τ : Real part, time decay / 実部，時定数
- ω : Imaginary part, frequency (oscillation) / 虚部，周波数

For discussing the frequency response, we suppose τ = 0 and analyze P(jω)
周波数応答を議論するために，τ = 0 として P(jω)を解析する
- | P(jω) | : Gain ゲイン
- arg P(jω): Phase 位相

$$P(s) = \frac{1}{K_e\left(T_m T_e s^2 + T_m s + 1\right)}$$

$$P(j\omega) = \frac{1}{K_e\left(T_m T_e \left(j\omega\right)^2 + T_m\left(j\omega\right) + 1\right)} = \frac{1}{K_e\left(1 - T_m T_e \omega^2 + T_m j\omega\right)} = a + jb$$

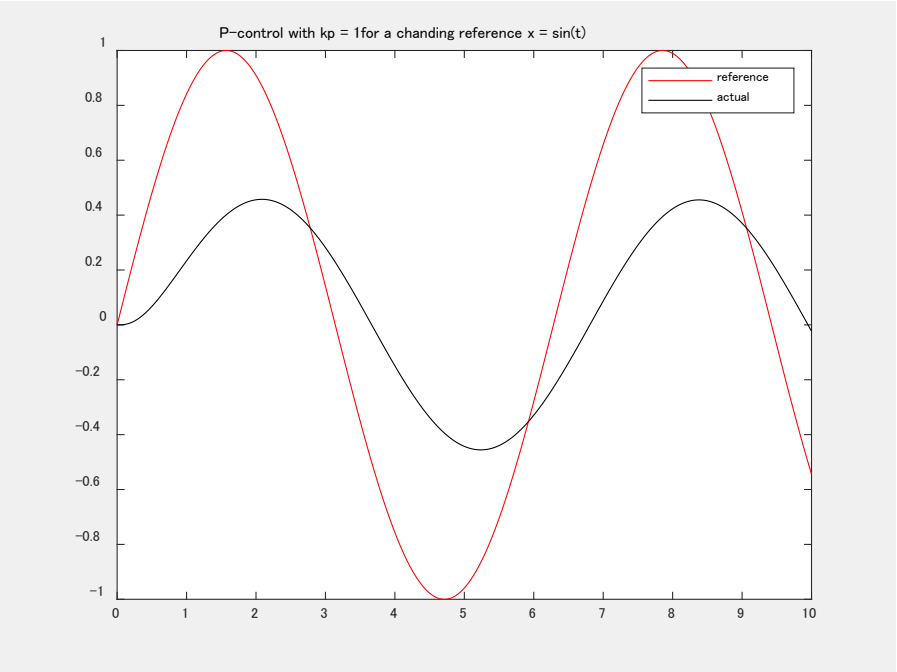
$$\left|P(j\omega)\right| = \sqrt{a^2 + b^2}$$

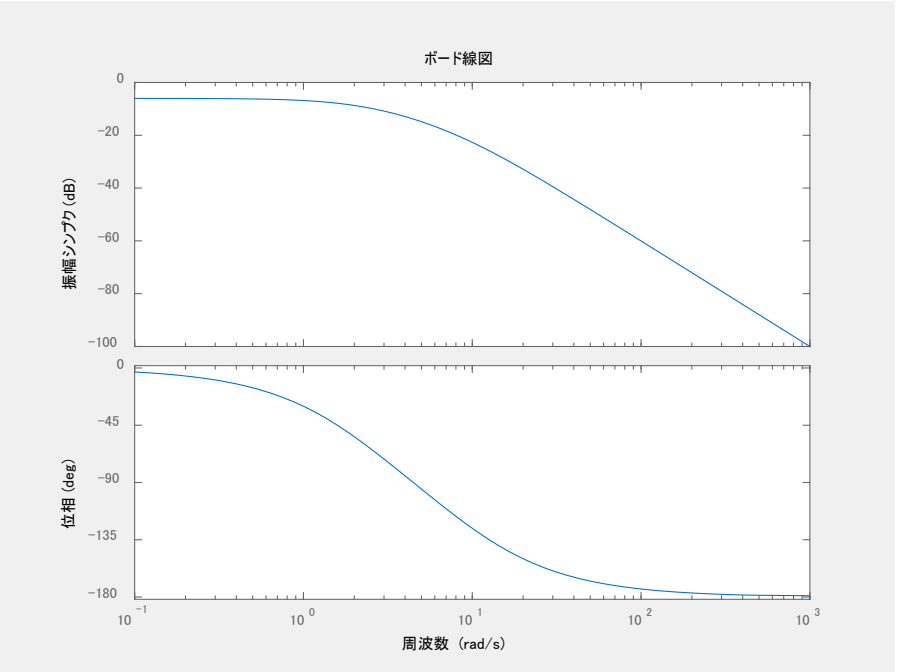
$$\arg P(j\omega) = \tan^{-1}\frac{b}{a}$$

CC BY

Bode Plot of P control with kp = 1 for DC motor speed

DCモータ速度のP制御のボード線図





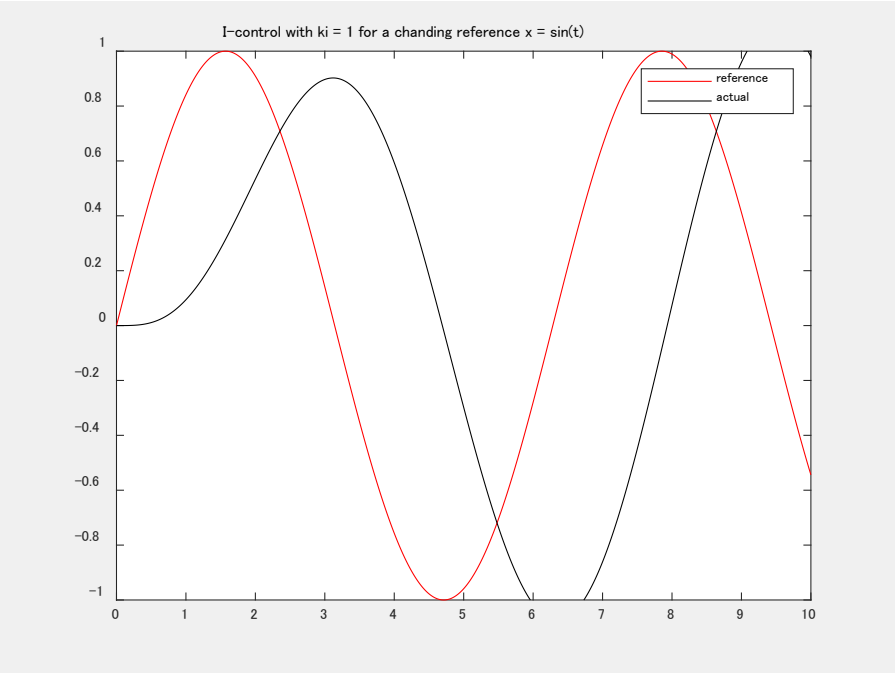
As the frequency goes higher, / 周波数が高くなるにつれ

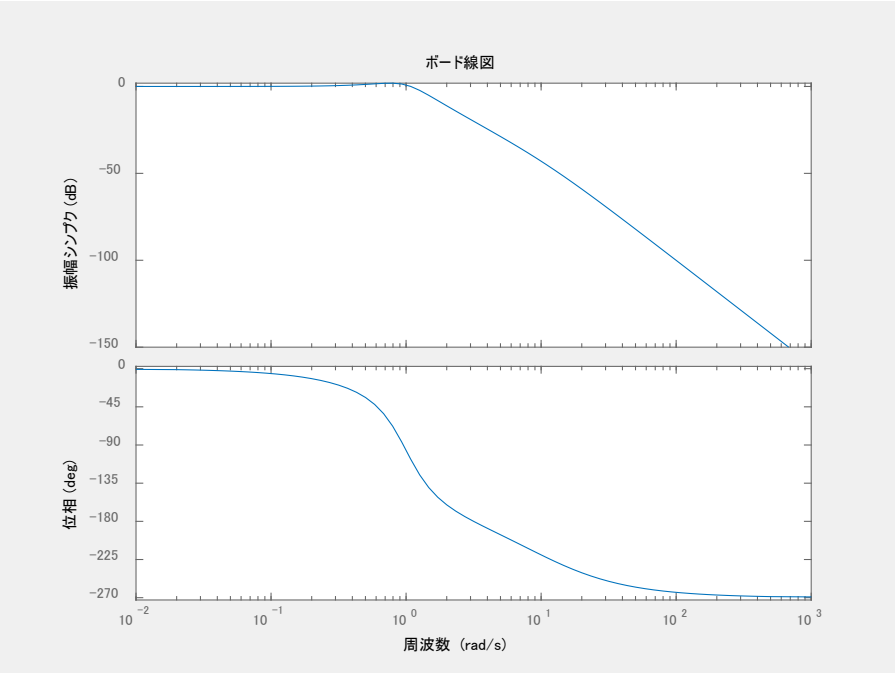
- the gain gets smaller / ゲインは小さくなり
- the phase changes from 0 to -180 (phase delay) / 位相は0から-180に変化する（位相遅れ）

CC BY

Bode Plot of I control with ki = 1 for DC motor speed

DCモータ速度のI制御のボード線図





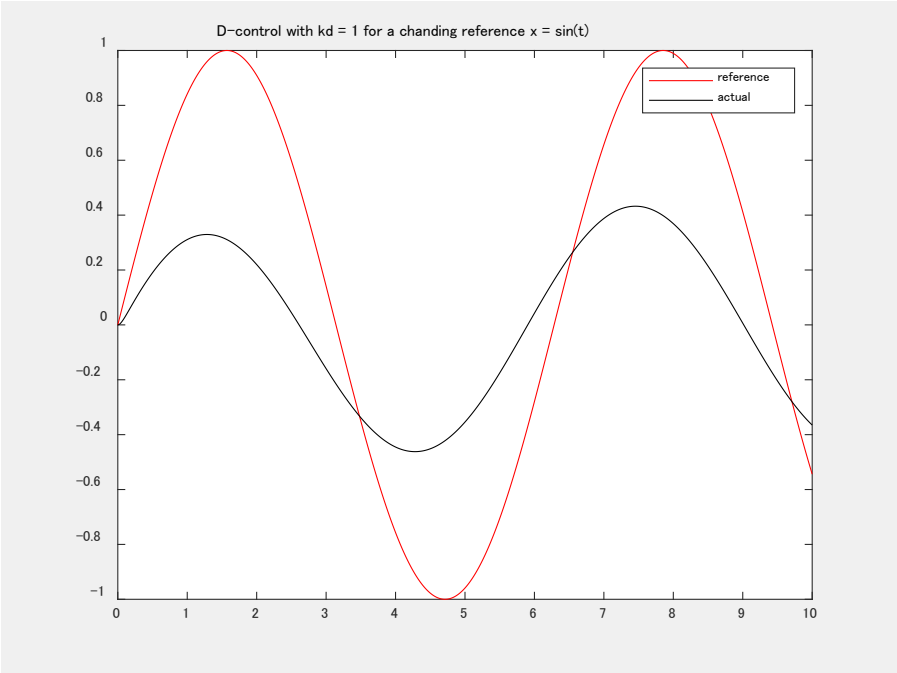
As the frequency goes higher, / 周波数が高くなるにつれ

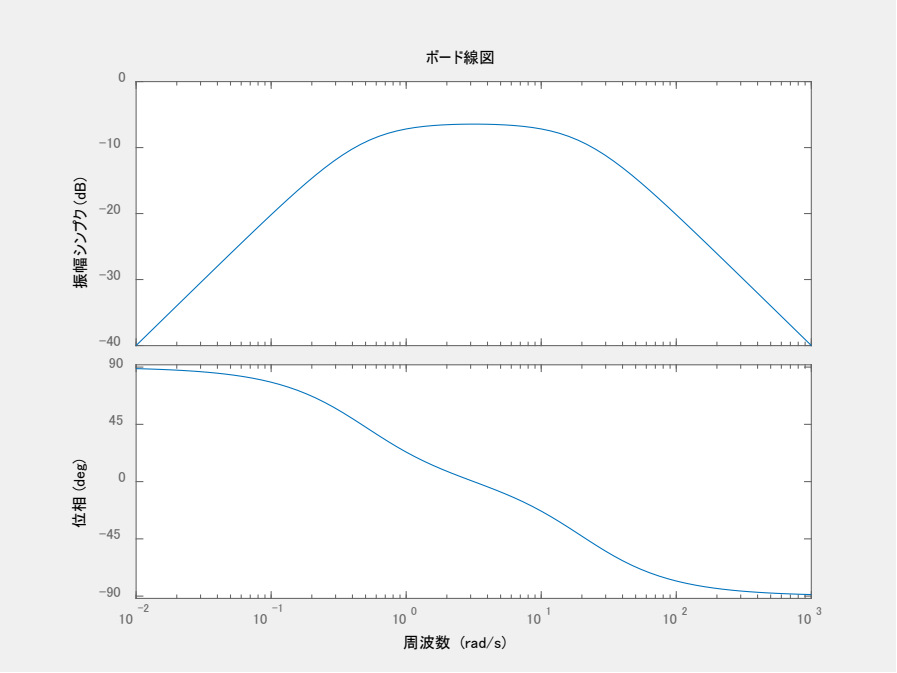
- the gain can have a peak, and gets smaller beyond it
  ゲインはピークを持ち，それ以降は減少する
- the phase changes from 0 to -270 (phase delay) / 位相は0から-270に変化する（位相遅れ）

CC BY

Bode Plot of D control with kd = 1 for DC motor speed

DCモータ速度のD制御のボード線図

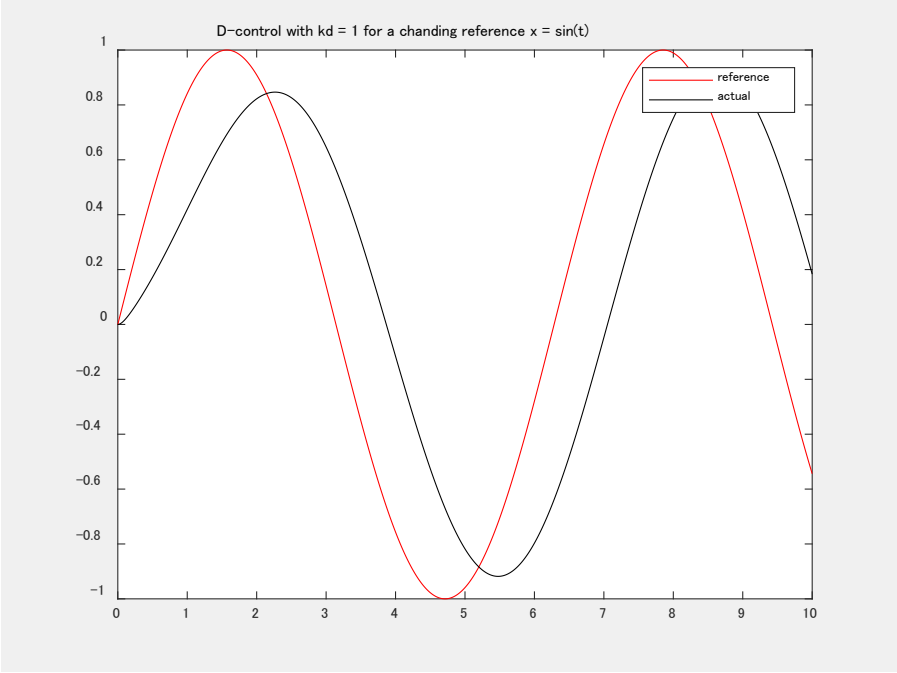


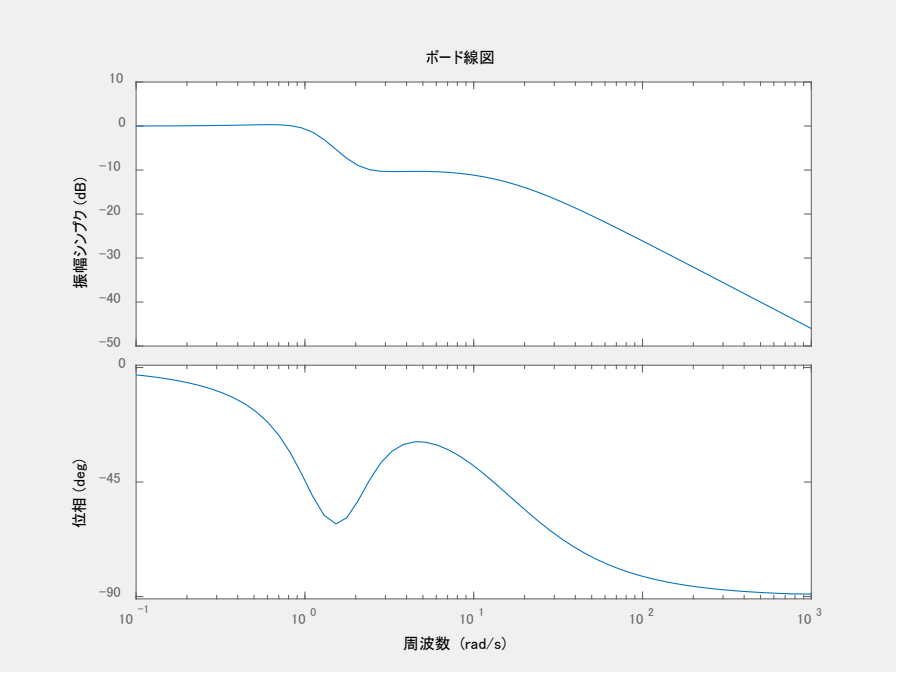


As the frequency goes higher, / 周波数が高くなるにつれ

- the gain start from small, increase gradually, have flat area, and decreases gradually
  ゲインは小さい値から始まり徐々に増加し，ほぼ一定の値を保った後，徐々に減少する
- the phase changes from +90 (phase lead) to -90 (phase delay)
  位相は90（位相進み）から-90に変化する

Bode Plot of PID control with kp = 1 ki = 2 kd = 0.5 for DC motor speed



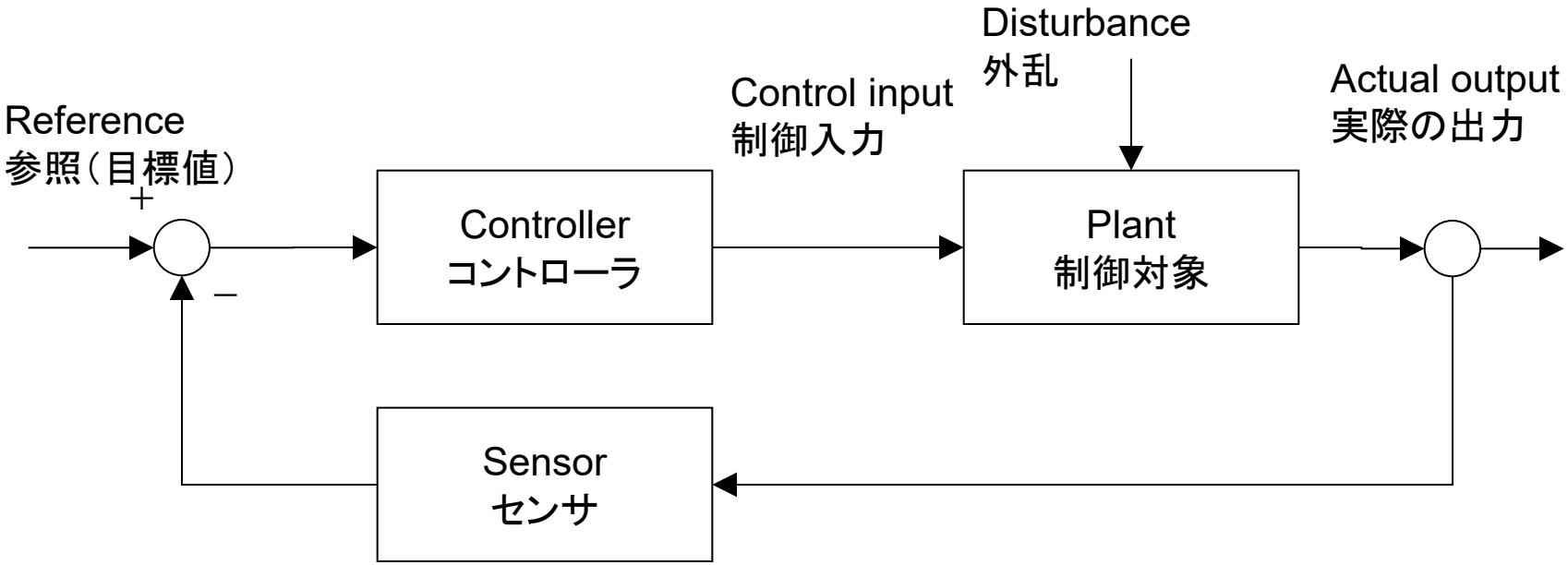


- By combining PID, the gain and phase can be improved.
  PIDを適切に組み合わせることで出力を改善することが可能
- This is PID gain tuning by the Bode plot
  これがボード線図によるPIDゲインの調整である.

Advantages of feedback control
フィードバック制御の利点

Feedback Control
フィードバック制御

- Use the difference between the reference and the actual output as online information
  参照（目標値）と現在値の差をオンライン情報として利用して制御する

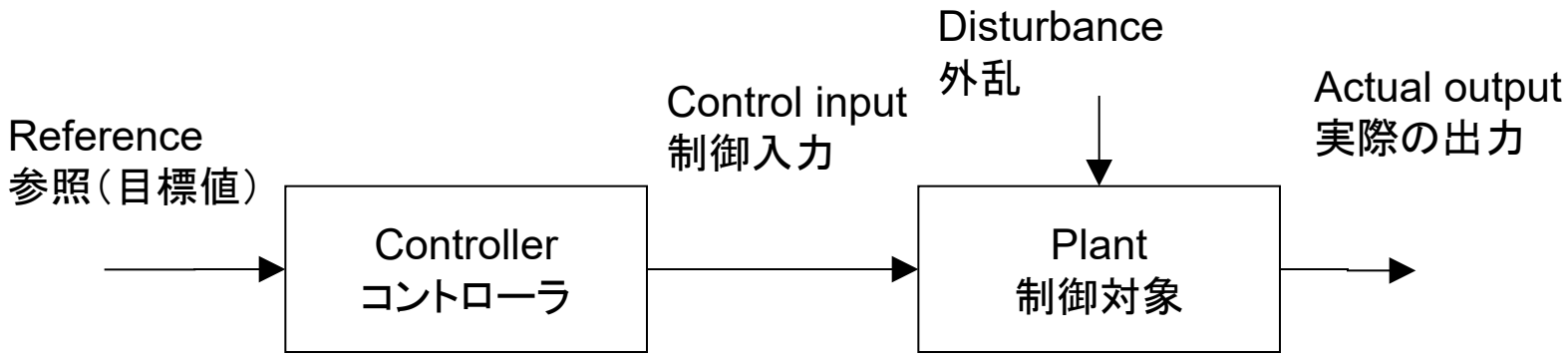


CC BY

Feedforward Control
フィードフォワード制御

- One shot control: It is effective if we know the characteristics of the plant (but it is impossible because we have disturbances and model changes)

- 単発型制御：制御対象の特性が完全に既知のときに有効（しかし実際には不可能，外乱やモデルの変化があるため）



CC BY

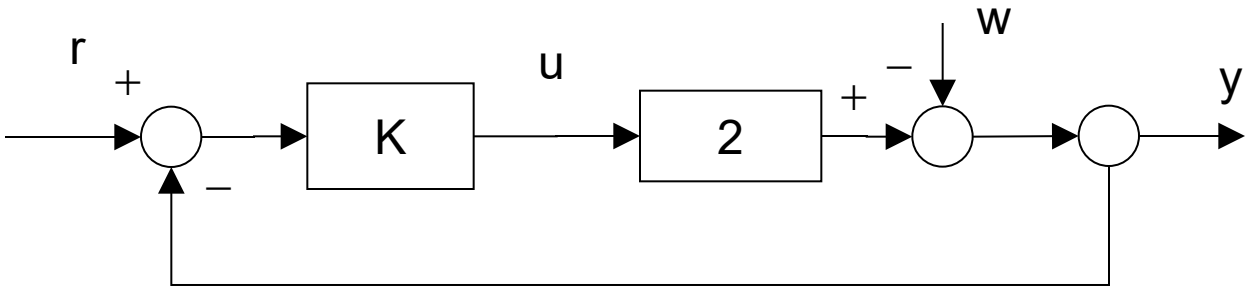
Advantages of Feedback Control
フィードバック制御の利点

- Stabilization of the plan / 制御対象の安定化
- Following to the references / 目標値の追従
- Suppression of the disturbances / 外乱の影響の抑制
- Suppression of the effects caused by the changes of the plant characteristics / 制御対象の特性変化の影響の抑制

CC BY

Speed Control of Submarine

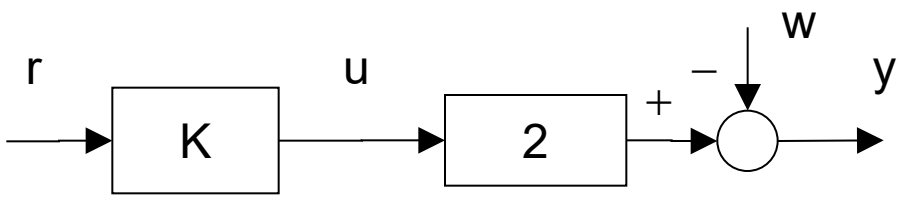
例：潜水艦の速度制御



- Simplified submarine / 単純化した潜水艦
  - r: Reference speed [m/s]目標速度
  - u: Motor current [A] モーター電流
  - y: Actual speed [m/s]　実際の速度
  - w: Water flow speed [m/s]水流

CC BY

Feedforward Control
フィードフォワード制御



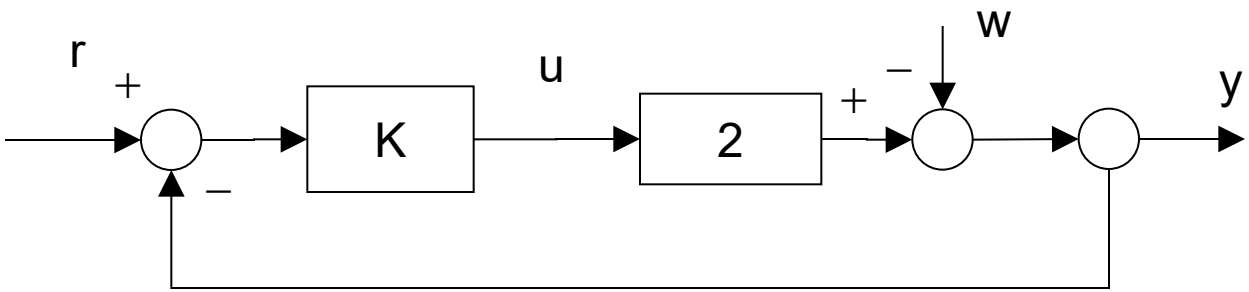
$$y = 2u - w \qquad y = 2Kr - w$$

$$u = Kr$$

If w = 0, for y=r

$$K = \frac{1}{2}$$

Feedback Control
フィードバック制御



$$y = 2u - w$$
$$u = K(r - y)$$

$$y = \frac{2K}{1+2K} r - \frac{1}{1+2K} w$$

If w=0, and K is large enough, e.g. K=100,
then y=r

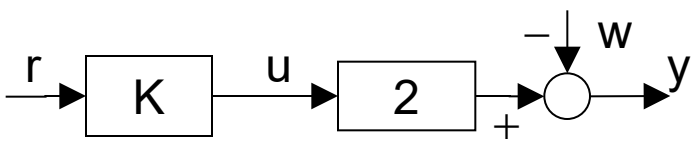
$$y = \frac{200}{201} r - \frac{1}{201} w$$

CC BY

Effect from Disturbances
外乱の影響

r=5 [m/s], w=2 [m/s]

Feedforward

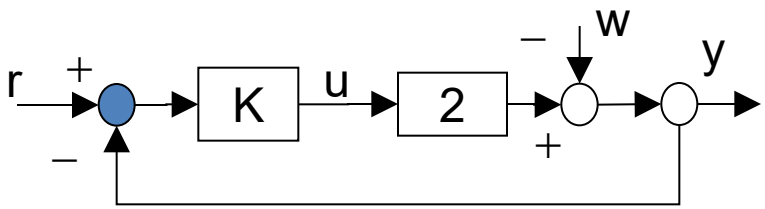


$$K = \frac{1}{2}$$

$$y = 2Kr - w$$
$$= 5 - 2$$
$$= 3$$

Error of 40%

Feedback



$$K = 100$$

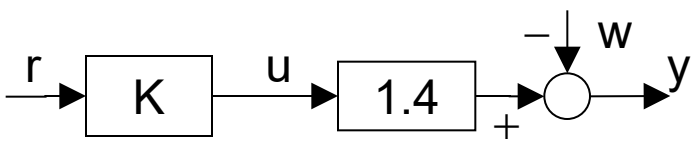
$$y = \frac{200}{201}r - \frac{1}{201}w$$
$$= \frac{200}{201}5 - \frac{1}{201}2$$
$$= 4.97$$

Error less than 1%

CC BY

Changes of Plant Characteristics
特性の変化

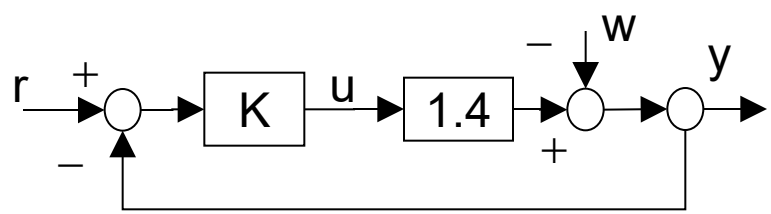
r=5 [m/s], w=0 [m/s], Plant: 2->1.4



$$K = \frac{1}{2}$$

$$y = 1.4Kr - w$$
$$= 0.7r$$
$$= 3.5$$

Error of 30%



$$K = 100$$

$$y = 1.4u = 1.4K(r - y)$$

$$y = \frac{1.4Kr}{(1 + 1.4K)}$$

$$y = \frac{140}{141}r = \frac{140}{141}5 = 4.96$$

Error less than 1%

CC BY

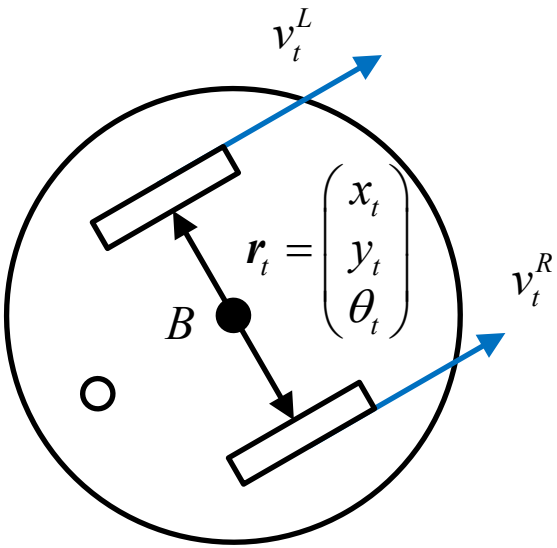
Summary of PID Control
PID制御のまとめ

- We have studies the concept of the feedback control
  フィードバック制御の概念を学んだ
- As the theory of the feedback control, we have studied the classical control theory, including
  フィードバック制御の理論として古典制御理論を学んだ
  - Transfer functions and Laplace transform / 伝達関数とラプラス変換
  - Block diagrams / ブロック線図
  - PID control / PID制御
- We have learned how to simulate DC motors and PID control systems in matlab
  DCモータとPID制御システムをmatalabシミュレートする方法を学んだ

Kinematics of Vehicles
車両の運動学

CC BY

Vehicle Kinematics 2Driving Wheel and1 Caster / 2DW1C車両の運動学



State /状態

$$\boldsymbol{r}_t = \begin{pmatrix} x_t \\ y_t \\ \theta_t \end{pmatrix}$$

Control / 制御

$$\boldsymbol{u}_t = \begin{pmatrix} v_t^L \\ v_t^R \end{pmatrix} \qquad \boldsymbol{v}_t = \begin{pmatrix} v_t \\ \omega_t \end{pmatrix} = \begin{pmatrix} \dfrac{v_t^L + v_t^R}{2} \\ \dfrac{v_t^R - v_t^L}{B} \end{pmatrix}$$

Vehicle Kinematics
車両の運動学

$$\dot{\boldsymbol{r}}_t = J(\theta_t)\boldsymbol{u}_t, J(\theta_t) = \begin{pmatrix} \dfrac{\cos\theta_t}{2} & \dfrac{\cos\theta_t}{2} \\ \dfrac{\sin\theta_t}{2} & \dfrac{\sin\theta_t}{2} \\ -\dfrac{1}{B} & \dfrac{1}{B} \end{pmatrix}$$

Odometer: Continuous system
オドメトリ: 連続系

$$\boldsymbol{r}_t = \int_0^t \dot{\boldsymbol{r}}_t dt = \int_0^t J(\theta_t)\boldsymbol{u}_t dt$$

Odometer: Discrete System
オドメトリ: 離散系

$$\begin{aligned} \boldsymbol{r}_k &= \boldsymbol{r}_{k-1} + \Delta t \dot{\boldsymbol{r}}_k \\ &= \boldsymbol{r}_{k-1} + \Delta t J(\theta_k)\boldsymbol{u}_k \end{aligned}$$

CC BY

Velocities of Vehicle and Wheels
車輪の速度と車両の速度

$$\begin{cases} v_t^L = r\omega_t^L \\ v_t^R = r\omega_t^R \end{cases}$$

$$\begin{cases} \omega_t^L : \text{Angular velocity of the left weel} \\ \omega_t^R : \text{Angular velocity of the right one} \\ \quad r : \text{the radius of the weels} \end{cases}$$

- $v^{L,R}$: the translation velocity of the left and right wheels / 左右の車輪の並進速度
- $\omega^{L,R}$: the angular velocity of them / 左右の車輪の回転速度
- r: the radius of them / 車輪の半径

$$\boldsymbol{v}_t = \begin{pmatrix} v_t \\ \omega_t \end{pmatrix} = \begin{pmatrix} \dfrac{v_t^L + v_t^R}{2} \\ \dfrac{v_t^R - v_t^L}{B} \end{pmatrix}$$

v: the translation velocity of the vehicle is the average of
ω: the angular velocity of the vehicle is the difference of
the velocities of the left and right wheels
左右の車輪速度の

- 平均が並進速度
- 差が回転速度

に対応する

CC BY

Update Positions from Velocities
速度から位置を更新

$$
\begin{pmatrix} x_{t+\Delta t} \\ y_{t+\Delta t} \\ \theta_{t+\Delta t} \end{pmatrix} = \begin{pmatrix} x_t \\ y_t \\ \theta_t \end{pmatrix} + \begin{pmatrix} \Delta t v_t \cos\theta_t \\ \Delta t v_t \sin\theta_t \\ \Delta t \omega_t \end{pmatrix}
$$

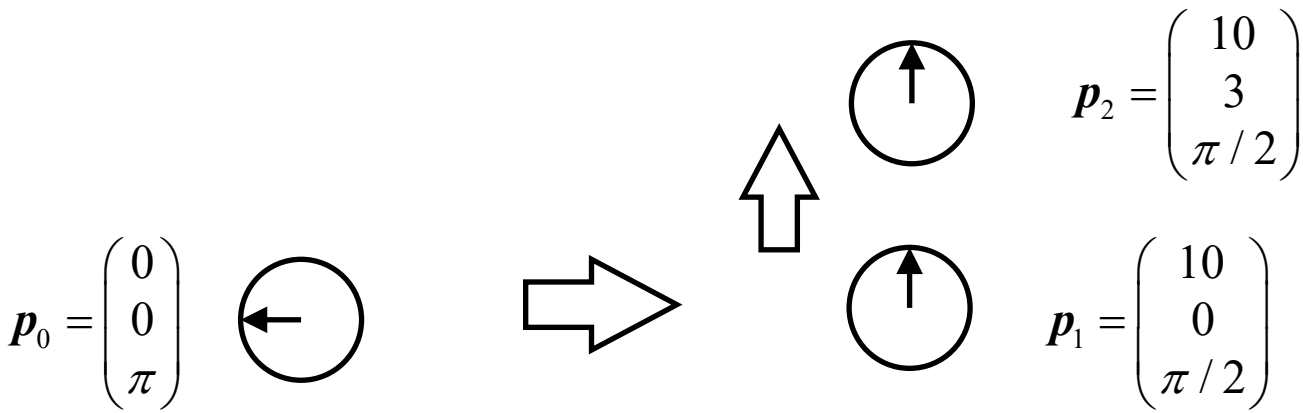
$$
= \begin{pmatrix} x_t \\ y_t \\ \theta_t \end{pmatrix} + \begin{pmatrix} \frac{v_t^R + v_t^L}{2} \Delta t \cos\theta_t \\ \frac{v_t^R + v_t^L}{2} \Delta t \sin\theta_t \\ \frac{v_t^R - v_t^L}{B} \end{pmatrix}
$$

CC BY

Path planning and velocity planning
経路計画と速度計画

Velocity planning
- Assume: a path is already planned
- Find(Plan): a sequence of robot translational and rotational velocity for a time period
- Find(Generate): a sequence of left and right wheel speed

速度計画
- 仮定：経路は計画済み
- 計画：各時間区間に対するロボットの並進速度と回転速度
- 生成：上に対応する車輪の回転速度



計画された経路の例/ Example of planned trajectory

Example of velocity planning
速度計画の例

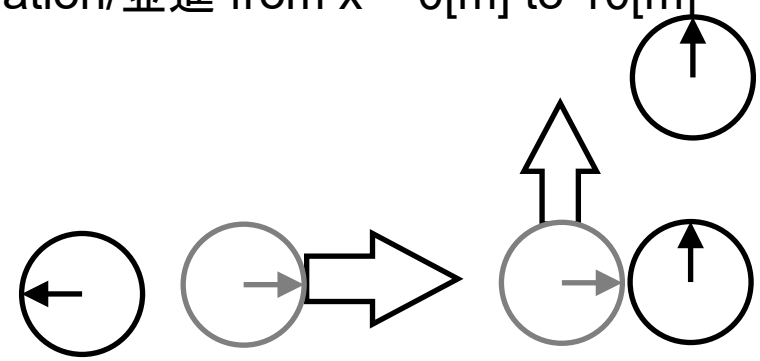
RT Style velocity planning
回転・並進型の速度計画

t = [12, 13] Rotation/回転from 0[deg] to pi/2 [rad]
w = pi/2[rad/s]
t = [13, 16] Translation/並進 y = 0[m] to 3[m]
v = 1[m/s]

t = [0, 2] Rotation/回転 from q= –pi[rad] to 0[rad]
w = pi/2[rad/s]
t = [2, 12] Translation/並進 from x = 0[m] to 10[m]
v = 1[m/s]

$$\boldsymbol{p}_2 = \begin{pmatrix} 10 \\ 3 \\ \pi/2 \end{pmatrix}$$

$$\boldsymbol{p}_0 = \begin{pmatrix} 0 \\ 0 \\ \pi \end{pmatrix}$$



$$\boldsymbol{p}_1 = \begin{pmatrix} 10 \\ 0 \\ \pi/2 \end{pmatrix}$$

CC BY

Example of wheel velocity planning
車輪速度生成の例

$$\boldsymbol{v}_t = \begin{pmatrix} v_t \\ \omega_t \end{pmatrix} = \begin{pmatrix} \dfrac{v_t^L + v_t^R}{2} \\ \dfrac{v_t^R - v_t^L}{B} \end{pmatrix}, B = 0.3$$

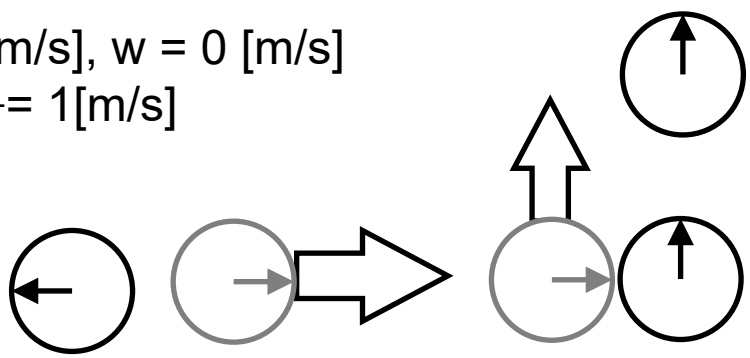
t = [12, 13]: v = 0[m/s], w = pi/2[deg/s]
-> v^R = 0.075pi [m/s], v^L= -0.075pi[m/s]
= 0.236[m/s]
t = [13, 16]: v = 1[m/s], w = 0[m/s]
-> v^R = 1[m/s], v^L= 1[m/s]

t = [0, 2]: v = 0[m/s], w = pi/2[rad/s]
-> v^R = 0.075pi [m/s], v^L= -0.075pi[m/s]

t = [2, 12]: v = 1[m/s], w = 0 [m/s]
-> v^R = 1[m/s], v^L= 1[m/s]

$$\boldsymbol{p}_2 = \begin{pmatrix} 10 \\ 3 \\ \pi/2 \end{pmatrix}$$

$$\boldsymbol{p}_0 = \begin{pmatrix} 0 \\ 0 \\ \pi \end{pmatrix}$$



$$\boldsymbol{p}_1 = \begin{pmatrix} 10 \\ 0 \\ \pi/2 \end{pmatrix}$$